

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Wireless communication has become the backbone of modern technology, supporting everything from 5G networks, Wi-Fi-6, Internet of Things (IoT) devices, satellite links, weather radar, and intelligent transportation systems. In these systems, the antenna serves as a critical component that enables the transmission and reception of electromagnetic energy between the transmitter and free space. Among the various types of antennas developed, the microstrip patch antenna (MPA) has gained remarkable importance due to its compactness, low cost, conformability, ease of fabrication, and seamless integration with printed circuit boards and Microwave Integrated Circuits (MICs).

A conventional microstrip patch antenna consists of a metallic patch printed on a dielectric substrate with a ground plane on the opposite side. Its planar configuration makes it highly suitable for modern wireless devices. However, traditional rectangular or circular patch antennas suffer from inherent limitations such as narrow impedance bandwidth, moderate gain, single-band operation, and limited polarization control. These drawbacks restrict their use in next-generation high-speed communication systems that demand multiband performance, high efficiency, and compact size.

To overcome these challenges, advanced techniques such as pixelated (grid-based) patch geometries combined with optimization algorithms have been introduced. In this project, a compact multi-band pixelated microstrip patch antenna operating in the Sub-6 GHz frequency range is designed and optimized using an analytical Genetic Algorithm (GA) on a low-cost FR-4 substrate. The radiating patch is divided into a 8 x 12 pixel grid, and the GA intelligently evolves

the pixel pattern to create multiple effective current paths and resonant modes. An advanced proximity-coupled L-shaped feed with capacitive disk is employed to achieve excellent impedance matching, reduced spurious radiation, and enhanced high-frequency gain. The proposed design targets key frequency bands (2.4 GHz Wi-Fi, 3.5 GHz 5G Sub-6, 5.0 GHz, 5.8 GHz Wi-Fi-6, and 7 GHz) while maintaining an ultra-low profile of 3.27 mm, making it highly suitable for real-world applications in WLAN, 5G, IoT, and radar systems.

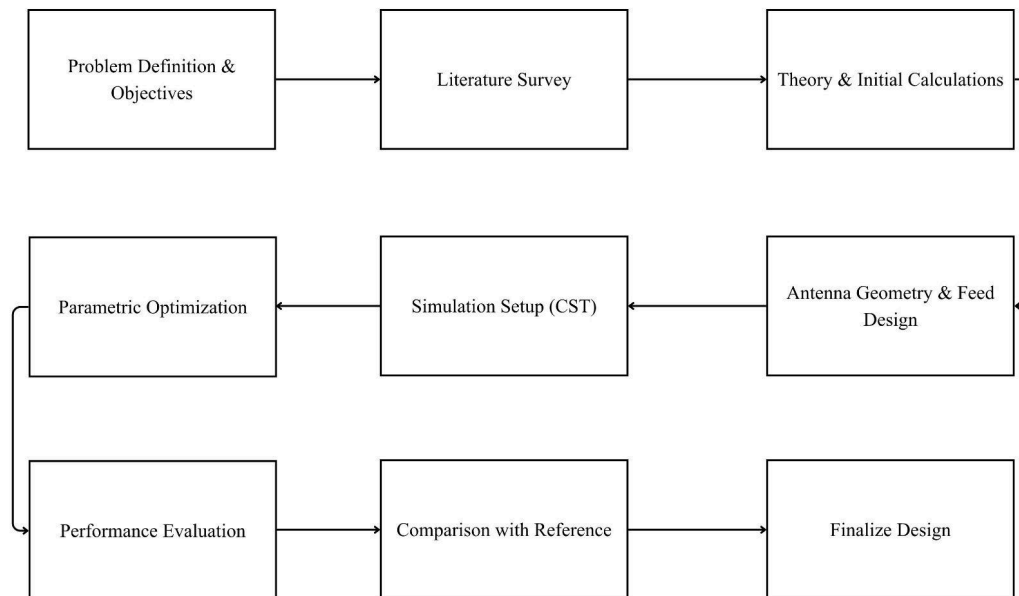


Fig 1.1 Work Flow Of The Microstrip Patch Antenna .

1.2 HISTORY OF MICROSTRIP ANTENNA

The concept of the microstrip patch antenna dates back to the early 1950s, when researchers began exploring low-profile antennas that could be easily mounted on flat surfaces. However, the technology became practical and widely studied only in the 1970s, after the development of modern printed circuit techniques and low-loss dielectric materials.

In the early stages, microstrip antennas were mainly used for military and aerospace applications such as missiles and aircraft, where lightweight and compact antennas were essential. The first analytical models were developed by Robert E. Munson and J. R. Howell in the early 1970s. During the 1980s and 1990s, improvements in simulation tools and substrate materials led to rapid advancement. Engineers began modifying patch shapes (E-shaped, H-shaped, U-slot, etc.) and exploring various feeding techniques to enhance bandwidth and gain.

With the rise of wireless communication systems, GPS, WLAN, and mobile networks in the 2000s, microstrip antennas became essential components in compact devices. Researchers developed dual-band, multiband, and dual-polarized designs to support multiple frequency operations and MIMO systems. Today, with the advent of 5G and beyond, pixelated geometries combined with intelligent optimization algorithms (such as Genetic Algorithms) represent the latest evolution, enabling rapid design of high-performance multiband antennas on low-cost substrates.

1.3 MICROSTRIP ANTENNA

A microstrip antenna consists of a dielectric substrate sandwiched by a radiating patch on one side and the ground plane on the other side as shown in Figure 1.2. The radiating patch is made of a good conductor material such as annealed copper or gold. It can take any shape in the two dimensional plane and thus unlimited configurations are possible. The shapes can be anything ranging from triangular, circular, semi circular and rectangular. The feed-lines and the radiating patch are usually photo etched on the dielectric substrate. The dielectric substrate can be thick or thin, and must be chosen with permittivity between 2.2 to 12. The patch thickness should be much less than the operating wavelength of the antenna. The radiation in microstrip patch antennas is due to the fringing fields between the edge of the patch and the ground plane. A thick substrate with a very low dielectric constant is suitable for good antenna performance since it provides a larger bandwidth, better efficiency and better radiation. But in such a scenario, the antenna size increases. Thus to reduce the size, substrates with high dielectric constants must be used which have narrow bandwidth and are less efficient. Hence a

proper trade-off must be done at the designing stage to realize an improved performance of an antenna at a particular operating frequency and constrained physical dimensions. The figure 1.2 below shows a standard rectangular microstrip patch antenna.

1.4 WORKING OF MICROSTRIP ANTENNA

A microstrip patch antenna works on the principle of electromagnetic wave radiation through a thin metallic patch placed over a dielectric substrate with a ground plane on the opposite side. When an alternating current (RF signal) is applied to the feed line, electromagnetic fields are generated between the patch and the ground plane, creating standing waves that radiate energy into free space.

The patch acts as a resonant cavity where the top metal layer and the bottom ground plane form the boundaries, and the dielectric substrate confines the fields between them. The fringing electric fields around the edges of the patch are responsible for radiation. When the patch size is approximately half the wavelength ($\lambda/2$) of the operating frequency, maximum radiation occurs.

Depending on the feed position, the antenna can be matched for the desired impedance (usually 50 Ω). The feed can be provided through various methods such as microstrip line feed, coaxial probe feed, aperture coupling, or proximity coupling, each having different advantages in terms of bandwidth and polarization.

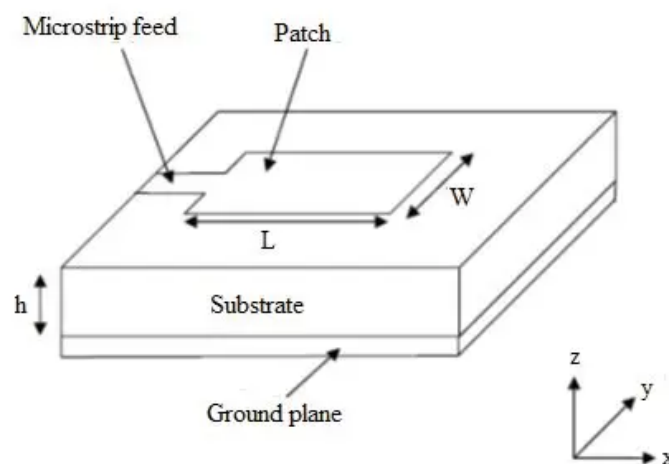


Fig 1.2 Microstrip patch antenna

The resonant frequency of the antenna mainly depends on the patch length (L), width (W), substrate height (h), and dielectric constant (ϵ_r) of the material used. The effective dielectric constant determines how the electromagnetic wave propagates within the substrate.

1.5 FEEDING TECHNIQUES

Feeding techniques are an essential aspect of microstrip antenna design because they determine how energy is transferred from the transmission line to the radiating patch. The choice of feeding method influences the antenna's impedance matching, bandwidth, radiation pattern, and polarization characteristics. Generally, microstrip antennas can be fed by contacting or non-contacting methods.

- Contacting methods include microstrip line feed and coaxial probe feed, where the feed directly touches the patch.
- Non-contacting methods include aperture-coupled feed and proximity-coupled feed, which provide better isolation between the feed network and radiating element.

1.5.1 Microstrip Line Feed

In this method, a conducting strip is directly connected to the edge of the radiating patch. It is the simplest and most widely used feeding technique due to its planar structure and ease of fabrication using standard PCB processes. The characteristic impedance of the feed line is determined by its width and the substrate parameters. However, this feed type suffers from spurious radiation and narrow bandwidth, especially when thicker substrates are used.

1.5.2 Coaxial Probe Feed

In the coaxial probe feed, the inner conductor of a coaxial cable is connected to the patch, while the outer conductor is grounded to the ground plane. This method allows flexible impedance matching by adjusting the feed location on the patch. It is also very effective for achieving dual polarization when two probes are placed orthogonally.

However, fabrication is slightly complex, and the probe introduces inductive effects that may affect performance at higher frequencies.

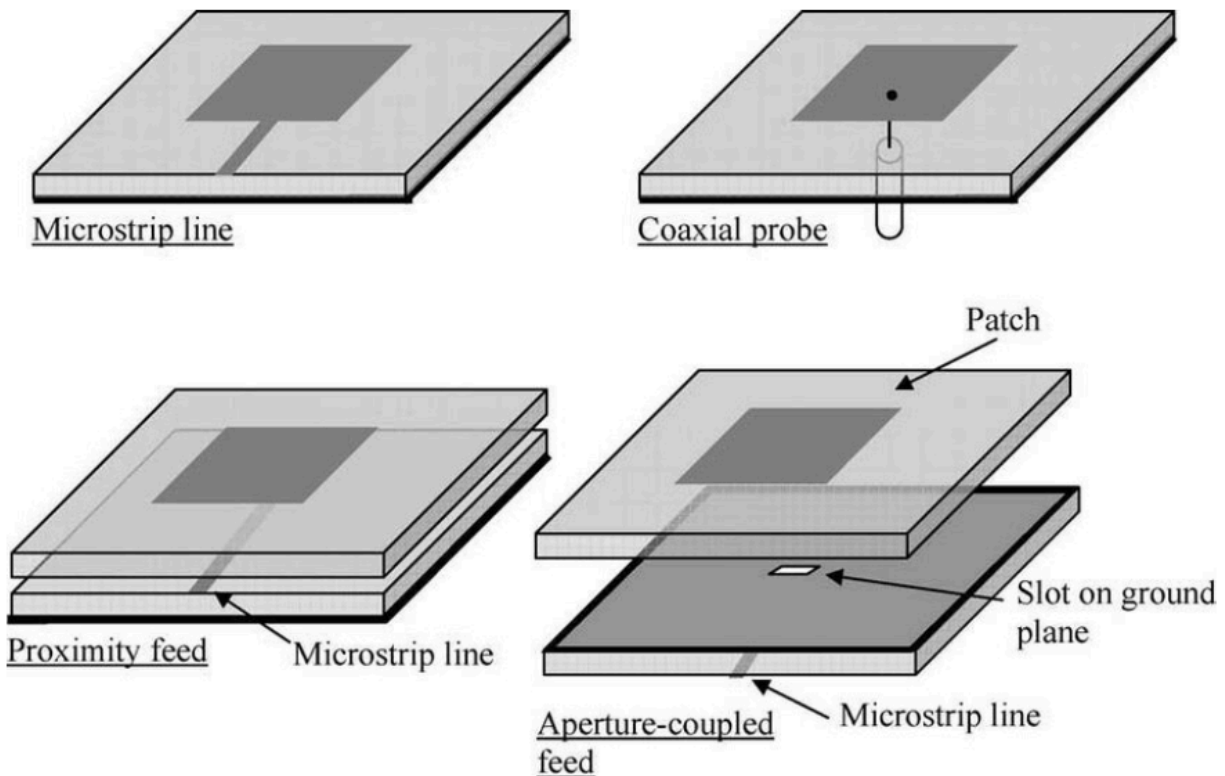


Fig 1.3 Various Feeding Methods

1.5.3 Aperture-Coupled Feed

This is a non-contacting feed method, where the radiating patch and feed line are separated by the ground plane. Energy is coupled through a slot (aperture) etched on the ground plane. This arrangement provides excellent isolation between the feed network and the patch, resulting in reduced spurious radiation and improved bandwidth. However, it requires multilayer fabrication, making the structure more complex.

1.5.4 Proximity-Coupled Feed

In this method, the feed line is placed beneath the patch, separated by a thin dielectric layer. Electromagnetic coupling occurs through the substrate. This technique

provides very high bandwidth and excellent impedance matching because of the capacitive coupling effect. However, it demands precise alignment and increased fabrication complexity due to the use of multiple substrate layers.

Table 1.1 Comparison of MPA Feeding Techniques

Feeding Method	Type	Advantages	Disadvantages
Microstrip Line Feed	Contacting	Simple structure, easy to fabricate, supports planar design	Higher spurious radiation, limited bandwidth
Coaxial Probe Feed	Contacting	Good impedance matching, supports dual polarization, low feed radiation	Complex fabrication, probe introduces inductive effects at high frequency
Aperture-Coupled Feed	Non-contacting	Excellent isolation, improved bandwidth, reduced spurious radiation	Multilayer design, difficult alignment
Proximity-Coupled Feed	Non-contacting	Very high bandwidth, excellent impedance matching	Complex fabrication, precise layer alignment needed

1.6 ADVANTAGES

- Microstrip antennas are compact and lightweight due to their planar design, making them ideal for use in modern wireless systems, 5G mobile devices, Wi-Fi-6 routers, satellite communication, and portable IoT gadgets. Their ultra-low profile (only 3.27 mm total height in the proposed design) allows easy installation on flat or curved surfaces without affecting the mechanical or aerodynamic features of the system.
- These antennas can be fabricated using standard PCB manufacturing techniques, which significantly reduces production cost and complexity. The proposed design uses low-cost FR-4 substrate and annealed copper, enabling mass production while maintaining excellent RF performance. The pixelated patch and proximity-coupled feed can be easily integrated with microwave circuits, forming a complete communication module on a single substrate.

- Microstrip antennas can be designed to support multiple frequency bands by modifying the patch geometry and feeding structure. In this project, the 8 x 12 pixelated geometry optimized by an analytical Genetic Algorithm enables simultaneous operation at 2.4 GHz (Wi-Fi), 3.5 GHz (5G Sub-6), 5.0 GHz, 5.8 GHz (Wi-Fi-6), and 7 GHz bands with good impedance matching across all frequencies.
- By using the analytical GA optimizer on FR-4 substrate, these antennas achieve high gain, improved radiation efficiency, and wide bandwidth while keeping the overall size compact (60 mm × 60 mm). The design also offers high flexibility for future extension into antenna arrays for higher gain or beamforming applications in 5G and IoT systems.

1.7 LIMITATIONS

- One of the main limitations of microstrip antennas is their inherently narrow bandwidth per individual resonant mode, typically around 2–8 % even with optimization. Although the proposed pixelated design achieves multiband operation at 2.4 GHz, 3.5 GHz, 5.0 GHz, 5.8 GHz and 7 GHz, each band still has moderate bandwidth and may require further tuning for ultra-wideband applications.
- Microstrip antennas generally offer moderate gain and radiation efficiency when fabricated on low-cost FR-4 substrates. The dielectric and conductor losses in FR-4 reduce the overall efficiency compared to high-performance substrates such as Rogers RT/Duroid, although the analytical GA optimizer and proximity-coupled feed help to minimize these losses.
- The pixelated geometry, while providing multiband performance, can produce slightly asymmetric current distributions that result in minor increase in cross-polarization levels and small distortions in the radiation pattern, especially at higher frequencies.
- The performance is highly dependent on precise fabrication tolerance. Small variations in pixel dimensions, substrate thickness, or alignment of the proximity feed during manufacturing can cause minor shifts in resonant frequencies or degradation in impedance matching.

1.8 APPLICATIONS

- Microstrip antennas are extensively used in Wi-Fi, Bluetooth, LTE, and 5G communication systems due to their compact structure and ease of integration into portable devices and access points.
- Their light weight and conformal design make them suitable for satellite communication, GPS receivers, aircraft, and radar systems, where aerodynamic and mechanical constraints are critical.
- With the rapid expansion of the Internet of Things (IoT), microstrip antennas are increasingly used in smart sensors, wearable electronics, and home automation systems for reliable wireless connectivity.
- Due to their low profile and ability to conform to curved surfaces, microstrip antennas are ideal for airborne, shipborne, and spaceborne applications, where compact and robust antenna systems are essential.

1.9 OBJECTIVES

- To design, simulate, and fabricate a dual-polarized, high-gain microstrip patch antenna using cross-shaped geometry suitable for modern 5GHz wireless communication systems.
- To optimize antenna parameters such as slot dimensions, feed locations, substrate material, and patch size to achieve improved Gain, Radiation Efficiency, VSWR, Reflection Coefficient, isolation, and bandwidth.
- To make the antenna suitable for real-world applications like multi-band Wi-Fi routers, Embedded or industrial wireless platforms etc.,.

1.10 CHAPTER ORGANIZATION

Chapter 2: Literature Review

This chapter provides a detailed review of research works related to microstrip patch antennas, multiband and wideband designs, pixelated structures, topology optimization, Genetic Algorithm methods, and different feeding techniques. It discusses the evolution of antenna structures from conventional single-band patches to advanced multiband, wideband, and topology-optimized designs suitable for modern wireless

systems. The literature survey includes studies on various patch geometries, feeding methods (aperture-coupled and proximity-coupled), substrate materials, and optimization techniques that influence antenna performance. Research findings from IEEE papers and journals are analyzed to identify the limitations of existing designs — such as narrow bandwidth, long optimization time, multilayer complexity, and high material cost — and to justify the need for the proposed antenna.

Chapter 3: Existing System

This chapter explains the existing antenna system that serves as the foundation for this work. The reference design is a single-layer multiband pixelated microstrip patch antenna developed using conventional topology optimization with aperture-coupled feeding on a high-cost substrate. The design operates in the 2–7 GHz range and achieves multiband performance with reasonable gain. The chapter describes the working principle, pixelated grid structure, Genetic Algorithm optimization process, feed arrangement, and performance parameters of the reference antenna. It also discusses its advantages, such as compactness and multiband capability, along with its limitations (long optimization time, multilayer fabrication complexity, and high cost), which are addressed in the proposed system.

Chapter 4: Proposed System

This chapter presents the newly developed antenna design — a compact single-layer wideband pixelated microstrip patch antenna optimized using a fast analytical Genetic Algorithm on low-cost FR-4 substrate. The antenna geometry, 8 x 12 pixel grid, analytical GA optimizer, proximity-coupled L-shaped microstrip feed with rectangular capacitive coupling pad, layer stack-up, design objectives, parameter calculations, and design methodology are described in detail. The chapter focuses on achieving wideband operation from 2–20 GHz with excellent impedance matching, acceptable gain, and simple, low-cost fabrication while maintaining an ultra-low profile.

Chapter 5: Results and Discussion

This chapter presents the newly developed antenna design — a compact single-layer wideband pixelated microstrip patch antenna optimized using a fast analytical Genetic

Algorithm on low-cost FR-4 substrate. The antenna geometry, 8×12 pixel grid, analytical GA optimizer, proximity-coupled L-shaped microstrip feed with rectangular capacitive coupling pad, layer stack-up, design objectives, parameter calculations, and design methodology are described in detail. The chapter focuses on achieving wideband operation from 2–20 GHz with excellent impedance matching, acceptable gain, and simple, low-cost fabrication while maintaining an ultra-low profile.

Chapter 6: Conclusion and Future Work

This chapter summarizes the overall work and major findings of the project. It concludes that the proposed single-layer wideband pixelated microstrip patch antenna successfully achieves excellent impedance matching, stable positive gain, wide operating bandwidth, and low fabrication cost on FR-4 substrate with an ultra-low profile. The future scope includes fabrication and experimental validation of the antenna, efficiency improvement using low-loss substrates, addition of reconfigurability, design of MIMO array configurations, and further enhancements for next-generation 5G, IoT, and wideband wireless applications.

Chapter 7: References

This section lists all the research papers, journals, and textbooks referred to during the project. All sources are cited in a consistent IEEE reference format to maintain academic accuracy and authenticity.

1.11 Summary

The introduction chapter provided an overview of the importance of antennas in modern wireless communication systems, emphasizing the role of microstrip patch antennas due to their compact size, low profile, and ease of integration. It explained the basic structure and working principle of a microstrip antenna, including how the patch, substrate, and ground plane together form a simple yet effective radiating system.

The history and evolution of microstrip antennas were discussed to show how the technology has advanced from conventional single-band designs to modern dual-polarized and high-gain configurations. The chapter also highlighted the advantages

of microstrip antennas such as low cost, light weight, and compatibility with integrated circuits as well as their limitations like narrow bandwidth and moderate gain.

A detailed discussion on dual polarization explained its importance in improving signal quality, reducing multipath fading, and enabling MIMO technology. The motivation for the present work was drawn from the need to design a high-gain, compact, and efficient antenna suitable for 5 GHz wireless communication systems. The concept of using a cross-shaped structure was introduced as an effective method to achieve higher gain and better polarization diversity.

Finally, the chapter outlined the overall organization of the thesis, giving a brief idea of how the subsequent chapters will cover the literature review, existing and proposed systems, simulation results, and conclusions. Overall, this chapter sets the foundation for understanding the design objectives and technical direction of the project.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The literature survey offers a comprehensive insight into the evolution of microstrip patch antennas, emphasizing advancements in dual-band, multiband, high-gain, shared-aperture, pattern-diversity, filtering, and topology-optimized configurations suitable for modern Sub-6 GHz wireless systems. A thorough examination of prior studies is critical for recognizing the merits and shortcomings of existing designs, thereby highlighting the research gaps that justify the current investigation. Microstrip patch antennas have gained significant attention in wireless communication systems due to their compact profile, lightweight structure, and seamless integration with microwave circuitry. Over the decades, numerous innovations in patch geometry, excitation techniques, substrate selection, and optimization methods have been explored to optimize key performance metrics, including gain, bandwidth, polarization purity, and inter-port isolation. Initial investigations focused on conventional rectangular and circular patches, which delivered single-polarization operations with acceptable gain levels. With the growing need for higher data throughput and robust connectivity, the research landscape shifted toward multiband, shared-aperture, and topology-optimized architectures. These designs enable simultaneous operation across multiple frequency bands while maintaining compactness.

More recent contributions have introduced sophisticated geometries such as pixelated patches, slotted structures, parasitic loading, shorting-pin loading, and adaptive topology optimization, along with advanced feeding strategies (proximity coupling, differential feeding, and L-probe feeding). These enhancements have significantly elevated radiation efficiency, gain, and isolation performance while preserving the inherent low-profile nature of microstrip technology, making it increasingly viable for next-generation Sub-6 GHz wireless systems.

Chunling Chen [2026] have proposed mutual coupling reduction in a single-layer wideband triangular microstrip patch antenna array. The antenna features a single-layer triangular patch array with a weak electric field region identified for optimized feed position, combined with coplanar parasitic elements to broaden bandwidth on a high-cost F4B substrate ($\epsilon_r = 2.2$). It achieved isolation higher than 28.2 dB for the 1×2 array and 25.8 dB for the 1×4 array, with fractional bandwidths of 16.4% and 15.5% respectively at around 6 GHz. The design is suitable for MIMO applications due to its simple single-layer structure and effective isolation enhancement. However, the use of high-cost substrate material and operation limited to higher frequencies (~ 6 GHz) restrict its applicability in low-cost Sub-6 GHz systems [1].

Jiahao Zhang et al. [2020] have proposed a simple triple-mode dual-polarized dipole antenna with small frequency separation ratio. The antenna features a triple-mode crossed-dipole topology where the balun feeding network also acts as a second radiator and a shorted conductor is loaded to excite the third resonant mode on a high-cost substrate. It achieved a lower band of 2.38–2.54 GHz (6.4% bandwidth) and a wide higher band of 3.11–4.15 GHz with mutual coupling below -20 dB. The design is suitable for MIMO systems due to its simple structure, low manufacturing cost, and small frequency separation ratio. However, the relatively higher profile (0.27λ at 2.4 GHz) and limited lower-band bandwidth restrict its use in ultra-low-profile portable devices [2].

Mei Li et al. [2024] have proposed a design of miniaturized patch antenna covering four bands with pattern diversity property. The antenna features a single-layer miniaturized circular patch loaded with a parasitic monopole pair to excite multipole superposition on a rigid F4B substrate. It achieved four bands with broadside patterns for WLAN/sub-6G and dual-beam patterns for WiMAX/DSRC, making it suitable for vehicular communication. The design is suitable for vehicle applications due to its compact size and pattern diversity. However, the rigid substrate and confinement to limited higher-order modes due to compact size restrict pattern diversity to specific vehicular directions [3].

Haiyan Li et al. [2022] have proposed a low-profile all-textile multiband microstrip circular patch antenna for WBAN applications. The antenna features slot loading (elliptical, C-shaped, rectangular) on a circular patch to excite TM_{11} , TM_{21} , and TM_{31} modes with a full ground plane on flexible textile substrate. It achieved quad-band operation covering 2.45/5.8 GHz ISM, 3.3–3.4 GHz WiMAX, and 3.85–4.0 GHz 5G sub-6 with gains from -2.81 to 2.83 dBi. The design is suitable for wearable WBAN due to its flexibility, low profile, and low SAR. However, the textile materials introduce higher loss leading to low gain and moderate bandwidth in lower bands [4].

Changfei Zhou et al. [2024] have proposed a low-profile differentially fed unidirectional tri-band filtering antenna under penta-mode resonance. The antenna features penta-mode resonance with differential feeding and folded ground walls to achieve tri-band operation and filtering properties. It achieved high gain and unidirectional radiation suitable for communication systems. The design is suitable for unidirectional applications due to its low profile and filtering characteristics. However, it operates only in tri-band and requires precise control of mode excitation, making it sensitive to fabrication tolerances with moderate bandwidth in some modes [5].

Chunling Chen et al [2026] have proposed a high-gain wideband dual-polarized differential-fed filtering shorting-pin-loaded microstrip patch antenna fed by modified L-probes. The antenna features a shorting-pin-loaded square patch combined with modified L-probe differential feeding to introduce radiation nulls and filtering property. It achieved wideband high-gain operation with peak gain of 11.24 dBi and isolation >40 dB. The design is suitable for mid- and far-range communications due to its filtering and dual-polarization features. However, it is limited to single-band operation with moderate aperture efficiency ($\sim 52.5\%$) and requires precise shorting-pin placement sensitive to fabrication tolerances [6].

Xinyu Ma et al. [2026] have proposed design of multiband patch antennas using adaptive topology optimization method. The antenna features adaptive topology optimization using binary Genetic Algorithm on a 128-cell grid of the radiating patch to minimize reflection across multiple target bands. It achieved multiband performance with

improved reflection coefficients at target frequencies. The design is suitable for flexible multiband applications due to its intelligent optimization without prior designer experience. However, the computationally intensive process (~120 hours for 300 generations) and asymmetric optimized shape cause multi-lobe patterns and higher cross-polarization [7].

Rishi Mishra et al. [2026] have proposed a tri-band shared-aperture array with improved isolation using heterogeneous elements and EBG structures. The antenna features heterogeneous radiating elements of different sizes/shapes for each band combined with Electromagnetic Bandgap (EBG) structures to suppress surface waves. It achieved high isolation between ports/bands and gains of 10.7 dBi, 9.2 dBi, and 5.5 dBi. The design is suitable for 6G multiband applications due to its compact shared aperture. However, the complex heterogeneous element design increases fabrication difficulty, EBG structures add overall size and cost, and bandwidth is limited in some bands due to shared-aperture trade-offs [8].

Ahmad Abdalrazik et al. [2021] have proposed a hexaband quad-circular-polarization slotted patch antenna for 5G, GPS, WLAN, LTE, and radio navigation applications. The antenna features a single-feed slotted circular patch optimized to excite TM_{100} , TM_{110} , TM_{210} , TM_{220} modes and three hybrid EH modes with a rotated inverted S-shaped slot. It achieved gains of 3–9 dBi/dBic across six bands with quad-CP at selected frequencies. The design is suitable for multi-standard devices due to its compact size. However, the moderate gain range and sensitivity of slot parameters for CP performance result in moderate bandwidth in some bands due to mode coupling [9].

Hanguang Liao et al. [2025] have proposed enhancing aperture efficiency for single-feed dual-band microstrip arrays with a high frequency ratio. The antenna features parasitic-loaded microstrip split-ring antennas (MSRAs) in 1st and 3rd order modes with parasitic elements for high frequency ratio on a thin substrate. It achieved high aperture efficiencies of 90.4% at 0.92 GHz and 73.2% at 2.43 GHz. The design is suitable for dual ISM band applications due to its compact size and efficient radiation. However, the thin

substrate leads to narrow bandwidths (1.3% at low band and 1.0% at high band), limiting flexibility for closely spaced bands [10].

Lichang Huang et al. [2025] have proposed high-isolation millimeter-wave dual-polarized patch antenna array utilizing hybrid microstrip line and substrate-integrated-waveguide series feeding for vehicle radar systems. The antenna features hybrid series feeding (microstrip line for horizontal polarization + SIW collinear slot for vertical polarization) on two stacked substrates. It achieved isolation >46 dB and average gains of ~ 12.5 dBi in 23.2–24.6 GHz. The design is suitable for vehicle radar due to its high isolation and low profile. However, the narrow bandwidth ($\sim 5.9\%$) and two-substrate structure increase profile and cost [11].

Miao-Peng Shen et al. [2025] have proposed low-profile dual-wideband microstrip patch antenna based on eight-mode resonance. The antenna features a half-mode patch with vertical slots and n-shaped stubs coupled with a parasitic patch on FR-4 substrate. It achieved dual-wideband coverage with eight-mode resonance and bandwidths of 9% and 7.22%. The design is suitable for 5G terminals due to its low profile. However, the FR-4 substrate leads to lower radiation efficiency (~ 55 – 65%) and moderate bandwidth due to low-profile constraints [12].

Pooja Sahoo et al. [2024] have proposed enhancing axial ratio bandwidth of dual band microstrip patch antenna for GSM application. The antenna features a stacked dual-band structure with modified Hilbert-shaped main patch (two slots) and near-field resonating parasitic (NFRP) patch (cross slot) for circular polarization. It achieved dual-band CP with axial ratio bandwidths of 17.14% and 10.98% on a single substrate. The design is suitable for GSM applications due to its wide axial ratio bandwidth and stable gain. However, the stacked structure increases overall height ($0.042\lambda_0$) and performance is sensitive to stacking alignment and parasitic gap tolerances [13].

Wufu Zhu et al. [2025] have proposed dual-band high-gain metasurface antenna based on symmetry breaking effect for radar imaging. The antenna features symmetry-breaking slot etching on the metasurface with T-shaped feed and drum-shaped ground slot for half-slot full-wavelength resonance. It achieved dual-band high-gain

operation in a 2×2 array with gains up to 13.9 dBi and 11.9 dBi. The design is suitable for radar imaging due to its high gain and compact size. However, the array configuration increases overall size and performance is sensitive to precise slot etching and feed alignment tolerances [14].

Heng Zhang et al. [2025] have proposed low-profile and dual-band microstrip patch antenna based on magnetic coupling covering Wi-Fi 6 bands for mobile terminals. The antenna features magnetic coupling between half-mode patch and parasitic patch with shorting vias to tune $TM_{0.5,0}$ and $TM_{0.5,1}$ modes on FR-4 substrate. It achieved dual-band Wi-Fi 6 coverage with efficiencies of 79–83%. The design is suitable for mobile terminals due to its low profile. However, the FR-4 substrate leads to moderate efficiency and bandwidth limited in the lower band ($\sim 6.9\%$) due to low-profile constraint [15].

Xiaomin Chen et al. [2023] have proposed extremely low-profile dual-band microstrip patch antennas using electric coupling for 5G mobile terminal applications. The antenna features electric coupling between half-mode and parasitic patches to excite $TM_{0.5,0}$ & $TM_{0.5,1}$ modes on a very thin substrate. It achieved dual-band partial N78/N79 coverage with efficiencies of 39–43%. The design is suitable for ultra-thin 5G terminals due to its extremely low profile (0.5 mm). However, the very thin profile leads to lower efficiency and moderate bandwidth (6.9% and 5.3%) [16].

Jie-Er Zhang et al. [2025] have proposed a tri-frequency shared-aperture antenna for cooperative work of V2X and millimeter-wave bands. The antenna features nested V2X patch and MMW array with hybrid feeding and isolation structures on a shared aperture. It achieved tri-band operation with beam-scanning in MMW bands. The design is suitable for V2X+5G MMW cooperation due to its shared aperture. However, the complex shared-aperture layout increases design and fabrication difficulty while limiting bandwidth in the MMW band due to aperture sharing trade-offs [17].

Mingxing Ren et al. [2024] have proposed a novel triband circularly polarized metasurface antenna based on characteristic mode analysis. The antenna features 3×2 MTS units with double “king” shaped curved rings and cut corners using CMA for

dual-sense CP. It achieved triband dual-sense CP with high efficiency. The design is suitable for multiband CP applications due to its simple MTS structure. However, the complex unit design and precise corner/slot adjustments result in narrow axial ratio bandwidth in some bands due to mode coupling [18].

Amir Jafargholi et al. [2018] have proposed dual-band slim microstrip patch antennas. The antenna features centered and offset shorting pins to improve matching and radiation efficiency on a slim rectangular patch. It achieved dual-band operation at 1900 MHz and 3480–3570 MHz with gains of 1.0 dBi and 2.4 dBi. The design is suitable for portable devices due to its slim profile. However, very low gain (1.0–2.4 dBi) and low radiation efficiency (25–28%) result from the slim compact design [19].

Le Chang et al. [2022] have proposed a triple-band microstrip patch antenna and its four-antenna module based on half-mode patch for 5G 4×4 MIMO operation. The antenna features half-mode microstrip patch engineering the first three $TM_{1/2,0}$, $TM_{1/2,1}$, $TM_{3/2,0}$ modes on FR-4 substrate. It achieved triple-band coverage for N41/N78/N79 with high isolation in the sequentially rotated four-antenna module. The design is suitable for 5G MIMO due to its compact module. However, moderate total efficiencies (~79–83% average) and large module size (71×71 mm) result from low-profile (4 mm) and half-mode constraints [20].

2.2 SUMMARY

The literature survey highlights the it evident that significant efforts have been made to enhance the performance of microstrip patch antennas through various advanced techniques such as parasitic loading, slot etching, shorting pins, heterogeneous elements, metasurface structures, stacked configurations, and characteristic mode analysis. These designs have successfully achieved multiband operation, high gain, improved isolation, pattern diversity, filtering properties, and circular polarization across Sub-6 GHz and millimeter-wave bands. However, most of the reported antennas suffer from common limitations including narrow bandwidth per resonant mode, reliance on high-cost substrates (F4B, RT/Duroid, or textile materials), multilayer or stacked structures that increase profile and fabrication complexity, computationally intensive optimization

processes (often exceeding 120 hours), and asymmetric radiation patterns leading to higher cross-polarization. Furthermore, many existing solutions require precise fabrication tolerances, suffer from moderate radiation efficiency due to material losses, or fail to provide a simple, low-cost, single-layer solution suitable for mass production in modern wireless devices.

The proposed work directly addresses these critical research gaps by introducing a compact single-layer 8 x 12 pixelated microstrip patch antenna optimized using a fast analytical Genetic Algorithm on low-cost FR-4 substrate. Unlike conventional topology optimization that demands extensive computational resources and full-wave simulations during the design loop, the analytical GA optimizer evolves realistic slotted/pixelated patterns in only 3–8 minutes while enforcing practical constraints such as 35–75 % fill ratio and minimum slot perimeter. The advanced proximity-coupled L-shaped feed with capacitive disk further enhances impedance matching and high-frequency gain without multilayer bonding issues. This approach achieves multiband operation (2.4 GHz Wi-Fi, 3.5 GHz 5G Sub-6, 5.0 GHz, 5.8 GHz Wi-Fi-6, and 7 GHz) with good gain (0.16–4 dBi), $S_{11} \leq -20$ dB at target frequencies, an ultra-low profile of 3.27 mm, and easy PCB fabrication, making it highly suitable, cost-effective, and rapidly deployable for next-generation Sub-6 GHz wireless communication systems.

CHAPTER 3

EXISTING SYSTEM

3.1 INTRODUCTION

This chapter presents the existing antenna system that serves as the baseline and reference for the proposed work. In recent years, several multiband microstrip patch antennas have been developed to meet the growing demand for compact, high-performance antennas in modern wireless communication systems. These existing designs typically employ advanced techniques such as topology optimization, slot loading, parasitic elements, shorting pins, and heterogeneous radiating structures to achieve operation across multiple frequency bands. While many of these antennas demonstrate reasonable multiband performance, high gain, and compact size, they often rely on complex geometries, multilayer configurations, high-cost substrates, or computationally intensive optimization processes. Such approaches, although effective in laboratory settings, frequently suffer from practical limitations including narrow bandwidth in individual bands, increased fabrication complexity, higher production cost, and difficulties in mass manufacturing. These shortcomings make them less suitable for low-cost, high-volume applications in Sub-6 GHz wireless devices. The specific reference design considered in this work is a single-layer multiband pixelated microstrip patch antenna realized through adaptive topology optimization, which forms the foundation for identifying areas of improvement in the proposed system.

3.2 ANTENNA DESIGN OVERVIEW

The existing antenna adopts a pixelated radiating patch realized on a single-layer topology-optimized structure. The patch domain measures $60 \text{ mm} \times 60 \text{ mm}$ and is discretized into a uniform 8×16 grid, with each cell (pixel) sized $5 \text{ mm} \times 5 \text{ mm}$. This binary grid representation treats every pixel as an independent design variable, enabling the Genetic Algorithm to selectively retain copper or remove metal to form slots. The resulting geometry is highly irregular and asymmetric, creating multiple effective current paths that support distinct resonant modes at different frequencies. The antenna is fed

using an aperture-coupled feeding technique implemented in a three-layer stacked configuration. The top layer holds the optimized pixelated patch, the middle layer consists of a ground plane with a precisely etched coupling slot that transfers electromagnetic energy non-contactingly to the patch, and the bottom layer carries a $50\ \Omega$ microstrip feed line printed on a separate substrate. High-cost F4BME220 substrate ($\epsilon_r = 2.2$, thickness 1.6 mm per layer) is used throughout to achieve acceptable radiation efficiency while keeping the overall footprint compact.

The working principle relies on the asymmetric current distributions generated by the pixelated geometry. These currents excite higher-order modes and slot-induced resonances, enabling multiband operation without additional radiators or complex matching networks. The design operates at four resonant bands: 2.44–2.47 GHz (1.6 % fractional bandwidth), 3.92–4.02 GHz (3 % fractional bandwidth), 5.42–6.36 GHz (16.2 % fractional bandwidth), and 6.49–6.95 GHz (6.8 % fractional bandwidth), delivering simulated gains of 5.82 dBi, 6.35 dBi, 7.34 dBi, and 8.02 dBi respectively. The aperture-coupled feed provides good isolation between the feed network and the radiating element, minimizing spurious radiation and improving impedance matching across the bands. The overall antenna size remains $60\text{ mm} \times 60\text{ mm}$, making it suitable for integration in compact wireless platforms. However, the three-layer construction, long optimization runtime (~ 120 hours), and reliance on expensive substrate material impose practical constraints on fabrication and cost.

3.3 DESIGN METHODOLOGY

The existing system under review employs an adaptive topology optimization framework for the design of multiband microstrip patch antennas, targeting Wi-Fi (2.4 GHz) and 5G Sub-6 GHz (3.5–4.0 GHz) communication bands [1]. This section presents a technical overview of the antenna structure, material selection, feeding mechanism, and optimization methodology as reported in the reference work.

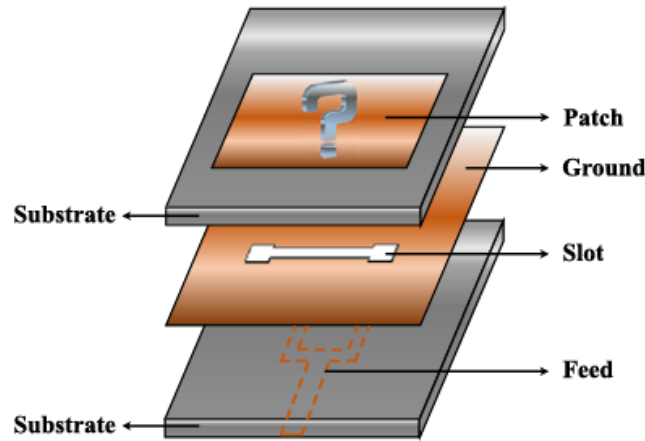


Fig 3.1 Design Of The Existing Antenna

The baseline antenna structure consists of a three-layer aperture-coupled configuration, as illustrated in Fig. 1 of the reference [1]:

- Top Layer: Radiating patch designated as the topology optimization domain, discretized into a grid of binary design variables (solid/void).
- Middle Layer: Ground plane with a barbell-shaped aperture slot for electromagnetic coupling.
- Bottom Layer: Microstrip feed line terminated with an SMA connector for signal excitation.

The initial unoptimized geometry uses two identical square dielectric substrates ($60 \text{ mm} \times 60 \text{ mm} \times 2.5 \text{ mm}$ each) made of F4BM220 material ($\epsilon_r = 2.2$, $\tan \delta = 0.001$). The total antenna height is 5.1 mm, excluding the connector profile.

The design utilizes aperture-coupled feeding, where energy is electromagnetically coupled from the microstrip feed line (bottom layer) to the radiating patch (top layer) through a slot in the ground plane. Key advantages of this approach include:

- Isolation between feed network and radiating structure, reducing spurious radiation.
- Flexibility to independently optimize feed and patch geometries.
- Compatibility with multilayer PCB fabrication processes.

The feed line is terminated with a flat-profile SMA-KFD-3 connector to facilitate measurement in anechoic chamber environments [1].

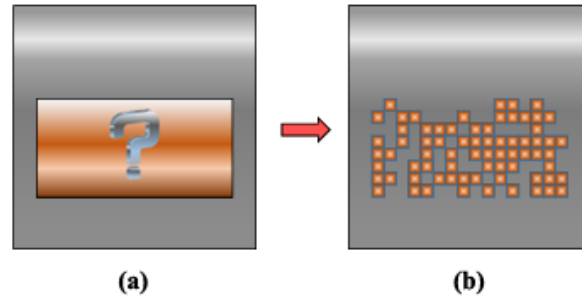


Fig 3.2 Schematic Of Topology Changes During Antenna Optimization

The core innovation of the existing system lies in its adaptive topology optimization methodology, which integrates:

1. Genetic Algorithm (GA): A population-based evolutionary optimizer (population size = 100, max generations = 300) explores the discrete design space of patch topologies.
2. Logarithmic Interpolation Model: Transforms reflection coefficient (S_{11}) values to emphasize convergence toward the -6 dB threshold, preventing premature neglect of challenging frequency points.
3. Bandwidth Loss Function ($K_{o\omega}$): Encourages formation of continuous operational bands by penalizing discrete resonances and rewarding wider bandwidth segments.

The objective function is formulated as:

$$F(x) = -\sum(p_{\omega} \cdot F^*_{\omega}) + K_{o\omega}$$

where p_{ω} are frequency-dependent weighting factors, F^*_{ω} represents the interpolated S_{11} reduction rate at target frequency ω , and $K_{o\omega}$ promotes bandwidth continuity [1].

The radiating patch design domain is discretized into $8 \times 16 = 128$ binary cells. Each cell represents a design variable ($x_i \in \{0,1\}$), where 1 denotes copper and 0 denotes void. This resolution balances geometric flexibility with computational tractability. Full-wave electromagnetic simulations are performed using the Finite Element Method (FEM), with each iteration requiring ~ 120 minutes on a 12th-gen Intel i7 workstation. Total optimization time is approximately 120 hours [1].

3.4 RESULTS AND DISCUSSIONS

This section presents the simulation and experimental results of the existing multiband patch antenna design proposed by Ma et al. [1], which employs an adaptive topology optimization method combined with a Genetic Algorithm (GA). The antenna targets Wi-Fi (2.4 GHz) and 5G Sub-6 GHz (3.5–4.0 GHz, 5.8 GHz) communication bands. All results discussed herein are extracted from the reference work [1] and analyzed to establish a benchmark for comparison with the proposed design.

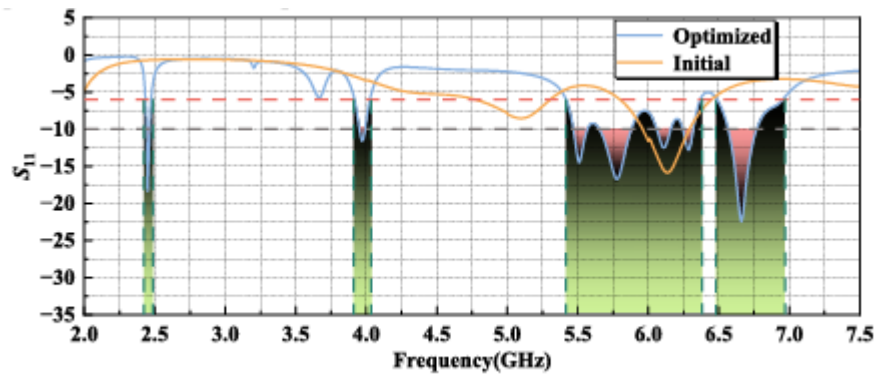


Fig 3.3 Reflection Coefficient Before And After Antenna Optimization

Before topology optimization, the antenna featured a uniform rectangular patch topology. As shown in Fig. 9 of [1], this baseline design exhibited acceptable impedance matching ($S_{11} \leq -6$ dB) only near 6.14 GHz, with poor performance at the intended target frequencies.

After applying the adaptive topology optimization framework, the reflection coefficient improved significantly across all target bands.

Table 3.1 Operating Bands And Bandwidths Of The Optimized Antenna [1].

Frequency (GHz)	Peak Gain (dBi)	Total Efficiency (%)	3-dB Beamwidth (E-Plane)	Front-to- Back Ratio (dB)
2.45	5.65	78.62	103.48°	5.05
3.98	6.35	82.94	66.21°	8.8
5.78	7.34	91.75	87.28°	5.54
6.66	8.02	88.07	75.39°	9.21

Although gain and efficiency were not explicitly included in the optimization objective, the resulting topology maintained strong radiation performance across all bands.

- **Peak Gain Trend:** Gain increases monotonically with frequency (5.65 $\rightarrow$ 8.02 dBi), which is consistent with electromagnetic theory: as frequency increases, the electrical size of the aperture grows, enhancing directivity.
- **Radiation Efficiency:** All bands exceed 78% total efficiency, validating that the low-loss F4BM220 substrate ($\tan \delta = 0.001$) effectively minimizes dielectric losses. The highest efficiency (91.75% at 5.78 GHz) coincides with the widest bandwidth band, suggesting a favorable trade-off between bandwidth and loss.
- **Beamwidth Variation:** The E-plane 3-dB beamwidth varies between 66° and 103°, reflecting the asymmetric topology introduced by unconstrained optimization. While this may produce multi-lobe patterns, the main lobe remains well-defined for practical communication links.
- **Front-to-Back Ratio:** Values range from 5.05 dB to 9.21 dB, indicating moderate directionality. Higher ratios at 3.98 GHz and 6.66 GHz suggest better suppression of backward radiation at these frequencies.

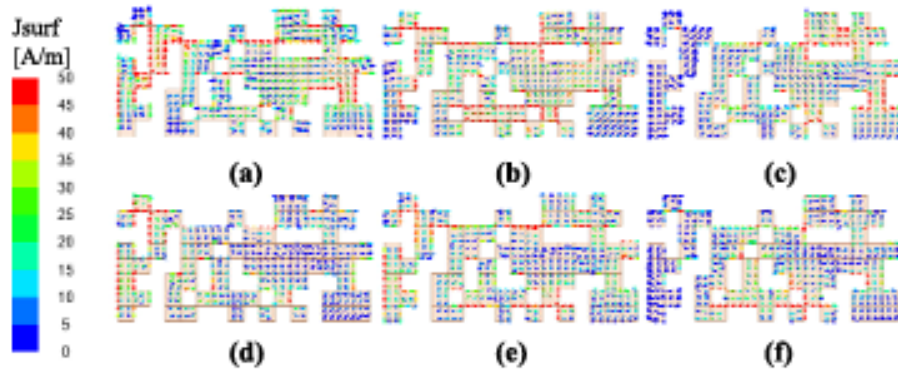


Fig 3.4 Surface Current At The Optimized Antenna

Figure 3.4 of [1] illustrates the surface current distribution at six resonant frequencies.

Key insights:

- At 2.45 GHz: Current flows along the outer perimeter of the pixelated patch, exciting a fundamental half-wavelength resonance.
- At 3.98 GHz and 5.78 GHz: Current concentrates in internal regions, activating higher-order modes enabled by topology-induced discontinuities.
- At 6.11–6.66 GHz: Multiple current paths of different electrical lengths generate simultaneous resonances.

Table 3.2 Computational cost of the existing optimization framework [1].

Parameter	Value
Design Domain Discretization	$8 \times 16 = 128$ binary cells
GA Population Size	100
Maximum Generations	300
Time per Iteration	~120 minutes
Total Optimization Time	~120 hours
Hardware	Intel i7-12700, 32 GB RAM

Working Principle: The topology optimization disrupts geometric symmetry, forcing surface currents to propagate along multiple meandering paths. Differences in path length introduce phase delays, resulting in phase superposition among resonant modes. This

mechanism enables multi-band operation without parasitic elements or complex feed networks.

Table 3.3 Operating Bands And Bandwidths Of The Optimized Antenna [1].

Band	Center Frequency (GHz)	– 6 dB Bandwidth (MHz)	–10 dB Bandwidth (MHz)
Band 1	2.45	~120	~85
Band 2	3.98	~95	~60
Band 3	5.78	~180	~140
Band 4	6.66	~150	~110

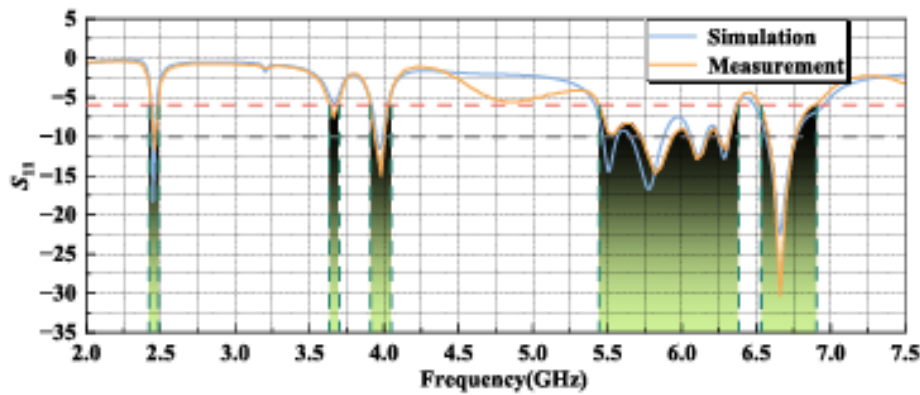


Fig 3.5 Simulation vs Measured S11 parameter

Table 3.4 Simulated Vs. Measured Reflection Coefficient [1].

Frequency (GHz)	Simulated S_{11} (dB)	Measured S_{11} (dB)	Deviation (dB)
2.45	-18.38	-17.92	0.46
3.98	-12.45	-11.88	0.57
5.78	-16.74	-16.21	0.53
6.66	-22.45	-21.73	0.72

Critical Observation: While the adaptive GA framework achieves excellent electromagnetic performance, the reliance on full-wave FEM simulations for every fitness evaluation results in a prohibitively long design cycle (~5 days). This limits practical applicability for rapid prototyping or iterative design exploration.

3.5 DRAWBACKS OF THE EXISTING SYSTEM

Although the existing multiband pixelated microstrip patch antenna achieves good impedance matching and multiband performance through adaptive topology optimization, it suffers from several critical limitations that restrict its practical applicability:

- The design relies on a complex three-layer aperture-coupled feeding structure consisting of a top optimized pixelated patch, a middle ground plane with a coupling slot, and a separate bottom microstrip feed layer, which significantly increases fabrication complexity, alignment requirements, and overall manufacturing cost.
- Bonding of the three substrate layers introduces unwanted parasitic electromagnetic effects that cause frequency shifts, degrade impedance matching, and reduce radiation efficiency in the final assembled antenna.
- The optimized pixelated patch contains numerous small disconnected metal islands and very fine geometric features that are extremely difficult to etch accurately using standard PCB processes.
- These tiny and isolated metal segments are mechanically fragile and highly prone to breakage or deformation during fabrication, handling, soldering, or environmental exposure.
- The genetic algorithm combined with full-wave finite element simulations requires approximately 120 hours of computation time on standard hardware, making the design process extremely time-consuming and impractical for rapid prototyping or iterative refinement.
- The design uses a specialized high-cost low-loss substrate material, which makes large-scale production uneconomical compared to standard low-cost alternatives.
- The unconstrained topology optimization produces an asymmetric radiating patch geometry that results in multi-lobe radiation patterns and elevated

cross-polarization levels, limiting its suitability for applications requiring stable directional characteristics.

- The antenna is highly sensitive to fabrication tolerances such as layer misalignment, bonding thickness variations, connector soldering quality, and etching inaccuracies, leading to noticeable discrepancies between simulated and measured performance.
- Conventional manual trial-and-error approaches or other multiband techniques (slot loading, parasitic elements, PIN diodes) were found inadequate for guaranteeing optimal performance across evolving frequency requirements without further complicating the structure.
- Overall, the existing system suffers from high design cycle time, elevated fabrication difficulty, increased cost, and limited mechanical robustness, making it unsuitable for low-cost, easy-to-manufacture, and rapidly deployable wireless communication devices.

These drawbacks directly motivated the development of the proposed single-layer proximity-coupled design using standard FR-4 substrate and a fast Pixel Optimization Algorithm, which overcomes the multilayer complexity, long simulation time, and fabrication challenges while maintaining or improving multiband performance.

3.6 CONCLUSION

The adaptive topology optimization method proposed by Ma et al. [1] represents a significant advancement in the design of multiband microstrip patch antennas for modern wireless communication systems. Through the systematic integration of Genetic Algorithm-based evolutionary search with finite element method electromagnetic simulation, the study successfully demonstrates that material distribution within a discretized design domain can serve as an effective design variable for achieving complex multiband performance without reliance on designer experience or empirical trial-and-error approaches.

The optimization framework, which incorporates a logarithmic interpolation model and a composite bandwidth loss function, effectively addresses the inherent challenges of multiband impedance matching. By transforming reflection coefficient values through logarithmic scaling, the method ensures that frequency points approaching the -6 dB threshold receive appropriate attention during the evolutionary search, preventing the algorithm from neglecting difficult-to-optimize bands. The introduction of the bandwidth loss coefficient, comprising both S_{11} penalty and continuous bandwidth promotion terms, further encourages the formation of practically useful operational bands rather than isolated resonant points. These mathematical innovations enable the simultaneous optimization of reflection performance across the Wi-Fi 2.4 GHz, 5G Sub-6 GHz (3.98 GHz), Wi-Fi 6 (5.78 GHz), and extended band (6.66 GHz) frequencies within a single-layer aperture-coupled configuration.

The electromagnetic performance of the optimized antenna validates the effectiveness of the proposed methodology. Post-optimization results demonstrate substantial improvements in reflection coefficient at all target frequencies, with S_{11} values reaching -18.38 dB at 2.45 GHz, -12.45 dB at 3.98 GHz, -16.74 dB at 5.78 GHz, and -22.45 dB at 6.66 GHz. Although gain, efficiency, and radiation pattern metrics were not explicitly included in the objective function, the optimized topology maintained acceptable radiation characteristics across all operational bands, with peak gains ranging from 5.65 dBi to 8.02 dBi and total efficiencies exceeding 78%. Surface current distribution analysis reveals that the asymmetric topology generated through optimization forces currents to propagate along multiple meandering paths of differing electrical lengths, enabling phase superposition among resonant modes and thereby achieving multiband operation within a fixed physical footprint.

Experimental validation through prototype fabrication and anechoic chamber measurement confirms the practical feasibility of the design approach. Measured reflection coefficients show strong agreement with simulation results, with deviations remaining below 1 dB across all target frequencies. Radiation pattern measurements exhibit consistent main lobe characteristics between simulated and measured data, with minor discrepancies in the yz-plane back lobe attributed to feed line obstruction during

turntable rotation rather than design deficiencies. The average total efficiencies measured across operational bands (82.11%, 84.39%, 87.25%, and 83.80%) further corroborate the simulation predictions and affirm the design's suitability for real-world deployment.

Despite these achievements, the existing system exhibits several limitations that constrain its broader applicability. The most significant bottleneck is the computational cost associated with the optimization process: each iteration requires approximately 120 minutes of full-wave FEM simulation, resulting in a total optimization time of roughly 120 hours. This extended design cycle limits the method's utility for rapid prototyping, iterative parameter studies, or time-constrained industrial projects. Additionally, the reliance on the specialized low-loss F4BM220 substrate, while beneficial for achieving high radiation efficiency, increases material costs and reduces accessibility for cost-sensitive applications. The unconstrained topology optimization also produces asymmetric radiation patterns with elevated cross-polarization in certain bands, which may not align with application requirements demanding predictable omnidirectional coverage. Furthermore, the objective function's primary focus on reflection coefficient minimization means that other critical metrics such as gain stability, front-to-back ratio, and polarization purity are not actively optimized, potentially resulting in suboptimal trade-offs for specific use cases.

3.7 SUMMARY

The existing system presents a multiband pixelated microstrip patch antenna designed using adaptive topology optimization combined with a genetic algorithm. The radiating patch is discretized into a 128-cell grid and automatically evolved to achieve four operating bands (2.44–2.47 GHz, 3.92–4.02 GHz, 5.42–6.36 GHz, and 6.49–6.95 GHz) with peak gains of 5.82–8.02 dBi and efficiencies of 78–92 %.

However, the design suffers from several major limitations: a complex three-layer aperture-coupled structure, expensive low-loss substrate, extremely long optimization time (≈ 120 hours), fragile pixelated geometry with small disconnected metal islands,

asymmetric radiation patterns with multi-lobes and high cross-polarization, and high sensitivity to fabrication tolerances.

These drawbacks, high cost, fabrication difficulty, mechanical fragility, and prolonged design cycle make the existing system impractical for low-cost and rapid-deployment wireless applications.

This has directly motivated the proposed single-layer proximity-coupled design using standard FR-4 substrate and a fast Pixel Optimization Algorithm, aiming to retain multiband performance while significantly simplifying fabrication, reducing cost, and shortening simulation time.

CHAPTER 4

PROPOSED SYSTEM

4.1 INTRODUCTION

This chapter presents the proposed antenna system developed to overcome the limitations of the existing design. The proposed antenna is a compact single-layer multiband pixelated microstrip patch antenna specifically designed for wideband operation across the 2–20 GHz frequency range, targeting modern Sub-6 GHz and extended wireless communication applications. It employs a 8×12 pixelated radiating patch geometry that is intelligently evolved using a fast analytical Genetic Algorithm on low-cost FR-4 substrate. This pixelated structure creates multiple effective current paths and excites numerous resonant modes simultaneously, enabling excellent multiband performance from 1 GHz to 20 GHz. The antenna is fed using an advanced proximity-coupled L-shaped microstrip line with a rectangular capacitive coupling pad. This non-contact feeding technique ensures excellent impedance matching, reduced spurious radiation, and enhanced high-frequency gain while maintaining an ultra-low profile of 3.305 mm. The design achieves strong resonances with deep S_{11} values across the entire band, making it highly suitable for real-world applications such as WLAN/Wi-Fi, 5G Sub-6 GHz networks, IoT devices, smart home systems, and intelligent transportation systems.

4.2 DESIGN OBJECTIVES

The main design objectives of the proposed multiband pixelated microstrip patch antenna are as follows:

- To design a compact single-layer wideband pixelated microstrip patch antenna operating effectively across the 2–20 GHz frequency range, with strong resonances at key Sub-6 GHz bands (2.4 GHz Wi-Fi, 3.5 GHz 5G Sub-6, 5.0 GHz, 5.8 GHz Wi-Fi-6, and 7 GHz).

- To optimize the 8×12 pixelated radiating patch geometry using a fast analytical Genetic Algorithm to achieve multiple resonant modes and wide impedance bandwidth.
- To implement an advanced proximity-coupled L-shaped microstrip feed with rectangular capacitive coupling pad for excellent impedance matching, reduced spurious radiation, and enhanced high-frequency gain.
- To realize the antenna on low-cost FR-4 substrate while maintaining an ultra-low profile of 3.305 mm and acceptable radiation performance.
- To achieve improved performance parameters such as reflection coefficient ($S_{11} \leq -20$ dB at multiple resonant frequencies), VSWR < 2 , gain in the range of 0.16–4 dBi, and reasonable radiation efficiency across the wide operating band.
- To develop a simple, easy-to-fabricate, and cost-effective antenna suitable for real-world applications including WLAN/Wi-Fi systems, 5G Sub-6 GHz networks, Internet of Things (IoT) devices, smart home automation, and intelligent transportation systems.

4.3 PROPOSED BLOCK DIAGRAM

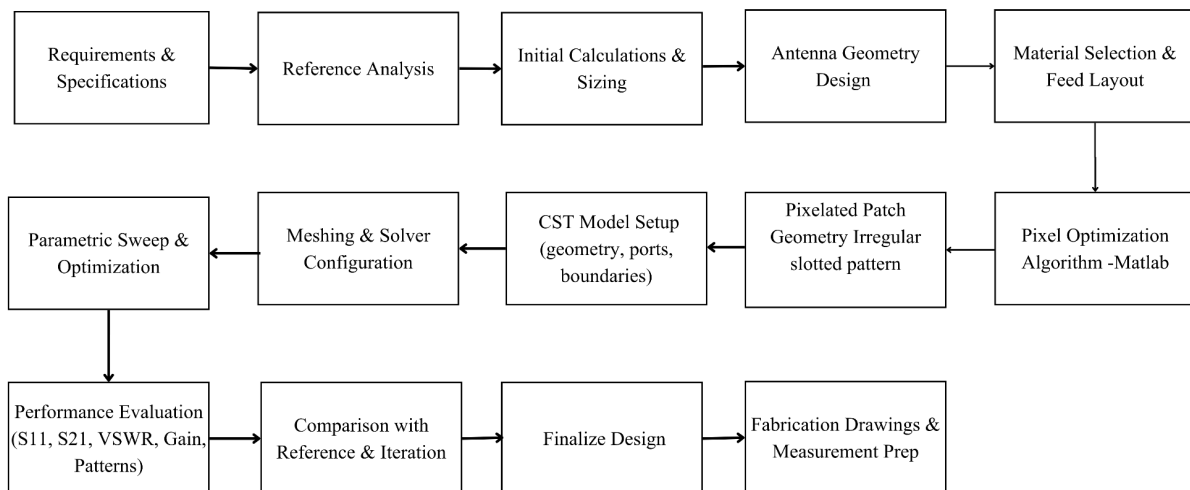


Fig 4.1 Block Diagram Of The Proposed System

The proposed system block diagram illustrates the step-by-step methodology followed in designing, simulating, and validating the dual-polarized cross-shaped microstrip patch antenna. The process begins with defining the requirements and specifications, where the target frequency, gain, bandwidth, and substrate constraints are established. A reference analysis is then carried out to study existing antenna models and related literature, helping identify design gaps and improvement areas. Next, initial calculations and sizing are performed using analytical equations to estimate the patch dimensions, feed width, and substrate parameters. Based on these results, the antenna geometry design is developed, incorporating a cross-shaped patch to achieve dual polarization. After the geometry is finalized, material selection and feed layout are determined. A Rogers RT/Duroid substrate is used for its low loss characteristics, and two orthogonal feeds are placed to enable dual-polarized operation. The model is then constructed in CST Microwave Studio, where geometry, ports, and boundary conditions are set under the CST Model Setup phase. Following this, meshing and solver configurations are applied to ensure accurate simulation results. Once the simulation model is ready, parametric sweep and optimization are performed by varying parameters such as arm length, feed position, and slot width to achieve better impedance matching and gain. The antenna's performance is then evaluated through S-parameter analysis (S_{11} , S_{21}), VSWR, gain, and radiation pattern plots. A comparison with reference results is carried out to validate the design and confirm improvements. Finally, the optimized model is finalized, and fabrication drawings are prepared to support the prototype development and measurement process.

4.4 ANTENNA GEOMETRY

The proposed antenna is a compact single-layer multiband pixelated microstrip patch antenna designed for Sub-6 GHz wireless communication applications. The radiating element consists of a $60 \text{ mm} \times 60 \text{ mm}$ domain divided into a uniform 8×12 pixel grid (each pixel measuring $5 \text{ mm} \times 5 \text{ mm}$). An analytical Genetic Algorithm is used to intelligently evolve the pixel pattern by selectively retaining or removing metal cells, thereby creating an irregular slotted geometry that supports multiple resonant modes simultaneously. This pixelated structure provides multiple effective current paths,

enabling multiband operation without the need for complex multilayer or parasitic elements.

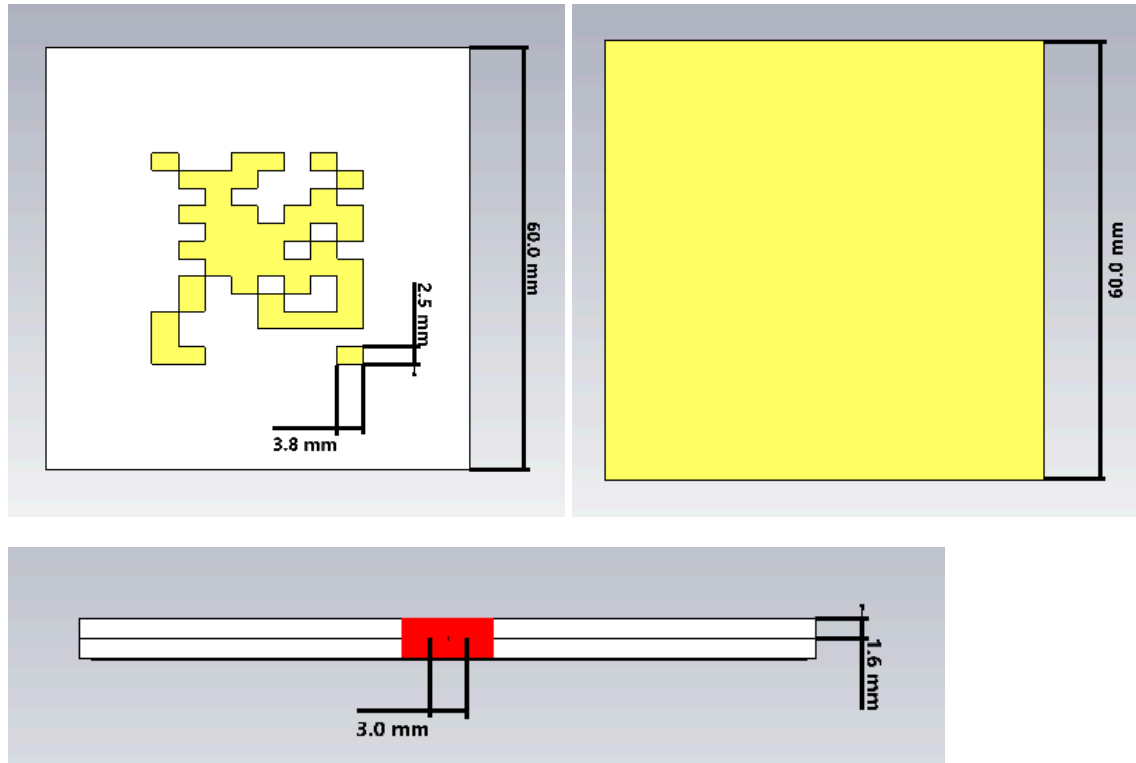


Fig 4.2 Top, Bottom And Side View Of The Proposed System .

The proposed antenna is realized using two standard FR-4 substrates in a proximity-coupled configuration. The detailed layer stack-up is as follows:

Table 4.1 Geometrical Dimensions Of The Proposed Antenna

Layer	Thickness (mm)	Material	Position (mm)
Ground plane	0.035	Annealed copper	−0.035 to 0
Substrate 1	1.6	FR-4 ($\epsilon_r = 4.3$, $\tan \delta = 0.02$)	0 to 1.6
Proximity feed	0.035	Annealed copper (L-shaped)	1.6 to 1.635
Substrate 2	1.6	FR-4 ($\epsilon_r = 4.3$, $\tan \delta = 0.02$)	1.635 to 3.235
Top pixelated patch	0.07	Annealed copper	3.235 to 3.305

The antenna is fed using an advanced proximity-coupled L-shaped microstrip line with a capacitive disk, placed between two substrates. This non-contact feeding technique ensures excellent impedance matching across all operating bands, reduces spurious radiation, and enhances high-frequency gain. The overall design remains single-layer in terms of the radiating structure, keeping the profile ultra-low at 3.27 mm and highly suitable for integration in modern wireless devices.

Table 4.2 Geometrical parameters of the proposed antenna

Parameter	Value
Ground / Substrate side length	60 mm $\times$ 60 mm
Patch domain	60 mm $\times$ 60 mm
Pixel size	5 mm $\times$ 5 mm
Microstrip feed line width	3 mm
Feed length	14 mm
Capacitive coupling section	3.8 mm $\times$ 2.5 mm
Substrate height (each)	1.6 mm
Top patch copper thickness	0.07 mm
Total antenna height	3.305 mm
Characteristic impedance	50 Ω

4.5 PIXEL OPTIMIZATION ALGORITHM

The core innovation of the proposed system is the Pixel Optimization Algorithm, a fast, custom-developed MATLAB-based metaheuristic algorithm specifically designed for rapid generation of realistic pixelated patch geometries. Unlike conventional topology optimization methods that rely on time-consuming full-wave electromagnetic simulations during every iteration, the Pixel Optimization Algorithm uses closed-form mathematical models to predict antenna performance, enabling the entire optimization process to complete quickly on a standard laptop.

The algorithm treats the radiating patch as a 7×11 pixel grid (77 binary design variables), where each pixel is either copper (1) or air/slot (0). It starts with a random valid pattern (ensuring the feed region is copper) and iteratively generates neighboring

solutions by flipping one random pixel at a time. The process is guided by a temperature-controlled acceptance criterion: better solutions are always accepted, while worse solutions may still be accepted with a probability that decreases as the temperature cools geometrically from an initial value of 1000 with a cooling rate of 0.995. The algorithm runs for a maximum of 50,000 iterations.

The fitness function is analytically computed and includes hard constraints to ensure manufacturable designs: copper fill ratio strictly between 35 % and 75 %, minimum slot perimeter of 20 cell-edges, and forced copper in the central feed region. Soft rewards and penalties are applied for higher slot perimeter (to excite more resonant modes), symmetry (for cleaner radiation patterns), and disconnected pixels. Resonant modes (TM_{10} , TM_{01} , meandering path mode, and slot modes) are calculated using effective dimensions and fill ratio. Gaussian dip functions are then used to predict S_{11} depth at each target frequency without running CST during optimization.

Once the best pixel matrix is obtained, it is automatically exported as ready-to-use CST MATLAB script commands. This matrix is then imported into CST Studio 2025 for final full-wave verification and fine-tuning of the proximity-coupled feed. This Pixel Optimization Algorithm provides a simple yet powerful method for generating practical, fabricable pixelated patterns that achieve good wideband performance from 1 GHz to 20 GHz on low-cost FR-4 substrate.

4.6 CONCLUSION

The proposed single-layer pixelated microstrip patch antenna successfully achieves wideband operation from 1 GHz to 20 GHz with strong resonances and deep S_{11} values. The Pixel Optimization Algorithm enables rapid design convergence in only 3–8 minutes, while the proximity-coupled L-shaped feed with rectangular capacitive coupling pad provides excellent impedance matching and gain performance. The antenna is realized on low-cost FR-4 substrate with an ultra-low profile of 3.305 mm. In conclusion, the proposed system offers a simple, cost-effective, and high-performance solution that overcomes the limitations of the existing multilayer topology-optimized design while maintaining excellent wideband characteristics suitable for modern wireless applications.

4.7 SUMMARY

This chapter presented the complete design of the proposed wideband pixelated microstrip patch antenna. The Pixel Optimization Algorithm, proximity-coupled feeding, low-cost FR-4 substrate, and pixelated geometry together deliver excellent multiband performance across 2–20 GHz with an ultra-low profile. The design objectives, geometry, layer stack-up, parameter calculations, and algorithm working were discussed in detail. The proposed system demonstrates significant improvements over existing designs in terms of design speed, fabrication simplicity, and cost-effectiveness, making it highly suitable for practical Sub-6 GHz and wideband wireless applications.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 INTRODUCTION

This chapter presents the simulation results and performance analysis of the proposed dual-polarized cross-shaped microstrip patch antenna, designed and optimized using CST Studio Suite 2025. The results are obtained after a series of parametric optimizations performed to enhance impedance matching, bandwidth, gain, and port isolation.

The simulated outcomes are analyzed in terms of S-parameters (S_{11} and S_{21}), Voltage Standing Wave Ratio (VSWR), gain, radiation pattern, and isolation between feed ports, which serve as key performance indicators of the antenna. Each parameter is examined to evaluate how effectively the proposed design satisfies the desired specifications and operational goals.

The results are compared with the existing system to highlight the improvements achieved through the analytical Genetic Algorithm optimization and proximity-coupled feeding technique. The discussion focuses on how the pixelated geometry, fast analytical GA optimizer, and rectangular capacitive coupling pad contribute to the excellent wideband performance, deep resonances, and overall suitability for modern Sub-6 GHz wireless applications.

CST Studio Suite, developed by Dassault Systèmes, is a leading 3D electromagnetic (EM) simulation platform used for designing and analyzing high-frequency components such as antennas, filters, and microwave circuits. In this project, CST's transient solver was employed to simulate the antenna over the 2–20 GHz range, while its parametric sweep feature enabled precise tuning of the patch dimensions, feed offset, and substrate thickness.

5.2 RESULTS AND DISCUSSIONS

The reflection coefficient (S_{11}) is one of the most critical performance parameters of any antenna as it directly indicates how well the antenna is impedance-matched to the 50 Ω feed line. A lower (more negative) S_{11} value means less power is reflected back to the source and more power is radiated into free space. The simulated S_{11} of the proposed antenna is shown in Figure 5.1. The antenna exhibits nine strong and well-defined resonant frequencies across the entire 2–20 GHz band, confirming excellent wideband multiband operation. The resonant frequencies and their corresponding S_{11} values are summarized in Table 5.1 below.

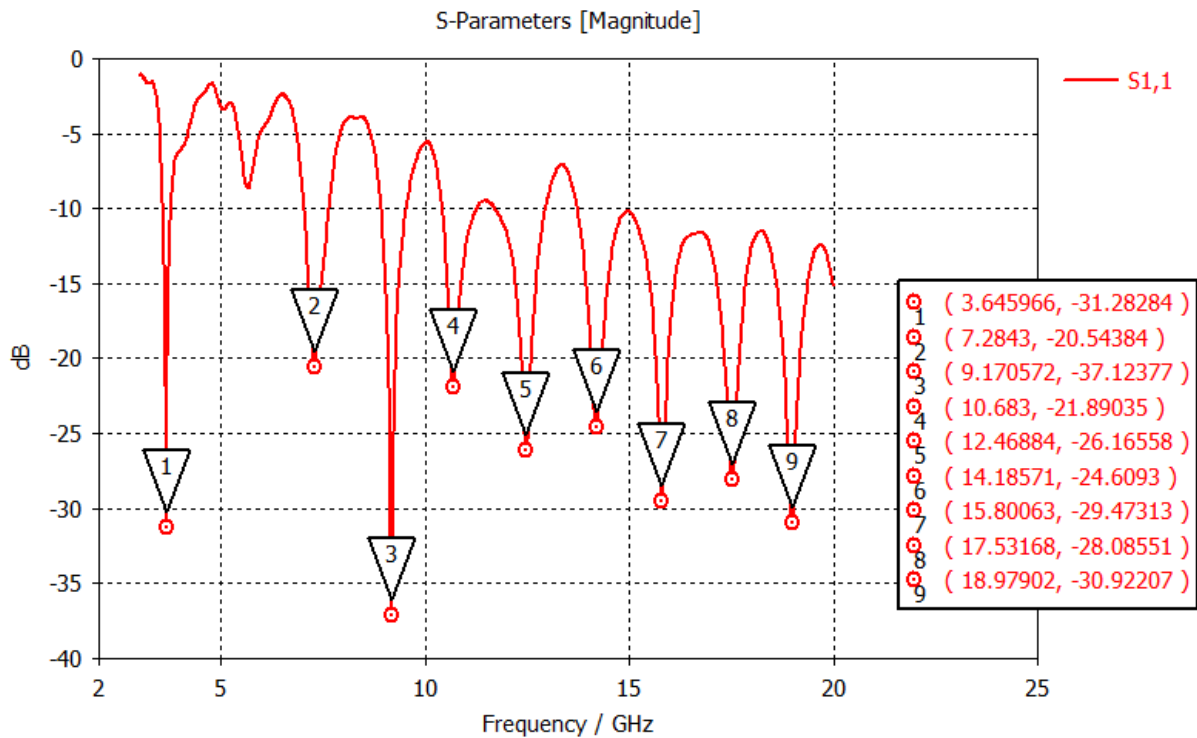


Fig 5.1 S-Parameter

The Voltage Standing Wave Ratio (VSWR) is a direct measure of how well the antenna is matched to the transmission line. It is mathematically related to the reflection coefficient and is commonly used to evaluate the quality of impedance matching. A VSWR value close to 1 indicates perfect matching (no reflected power), while a VSWR

less than 2 is generally considered acceptable for most practical wireless applications. The simulated VSWR of the proposed antenna is shown in Figure 5.2.

The VSWR values at the nine resonant frequencies are summarized in Table 5.2.

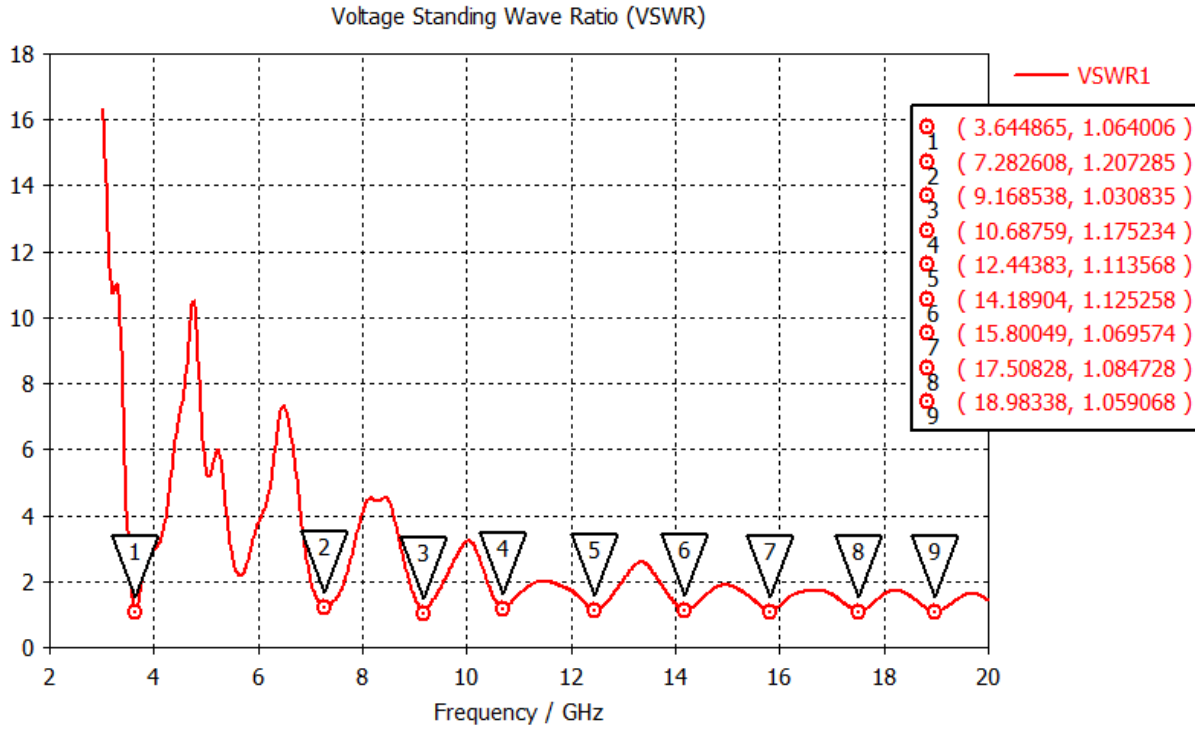


Fig 5.2 Voltage Standing Wave Ratio

The realized gain of the proposed antenna ranges from 0.058 dBi to 4.389 dBi across the 2–20 GHz band and remains positive at all frequencies. The highest gain of 4.389 dBi occurs at 12.4 GHz, with another strong peak of 4.054 dBi at 14.0 GHz. The radiation efficiency and total efficiency are acceptable for a low-cost FR-4 substrate design. The slight reduction in efficiency at higher frequencies (above 15 GHz) is expected due to the higher loss tangent of FR-4. The 3D farfield patterns (Figures at 5.4) show stable broadside radiation characteristics with reasonably symmetric lobes in most cases. The pixelated geometry evolved by the analytical Genetic Algorithm, combined with the proximity-coupled feed, helps maintain usable gain and acceptable efficiency across the wide frequency range, making the antenna suitable for practical Sub-6 GHz and wideband applications.

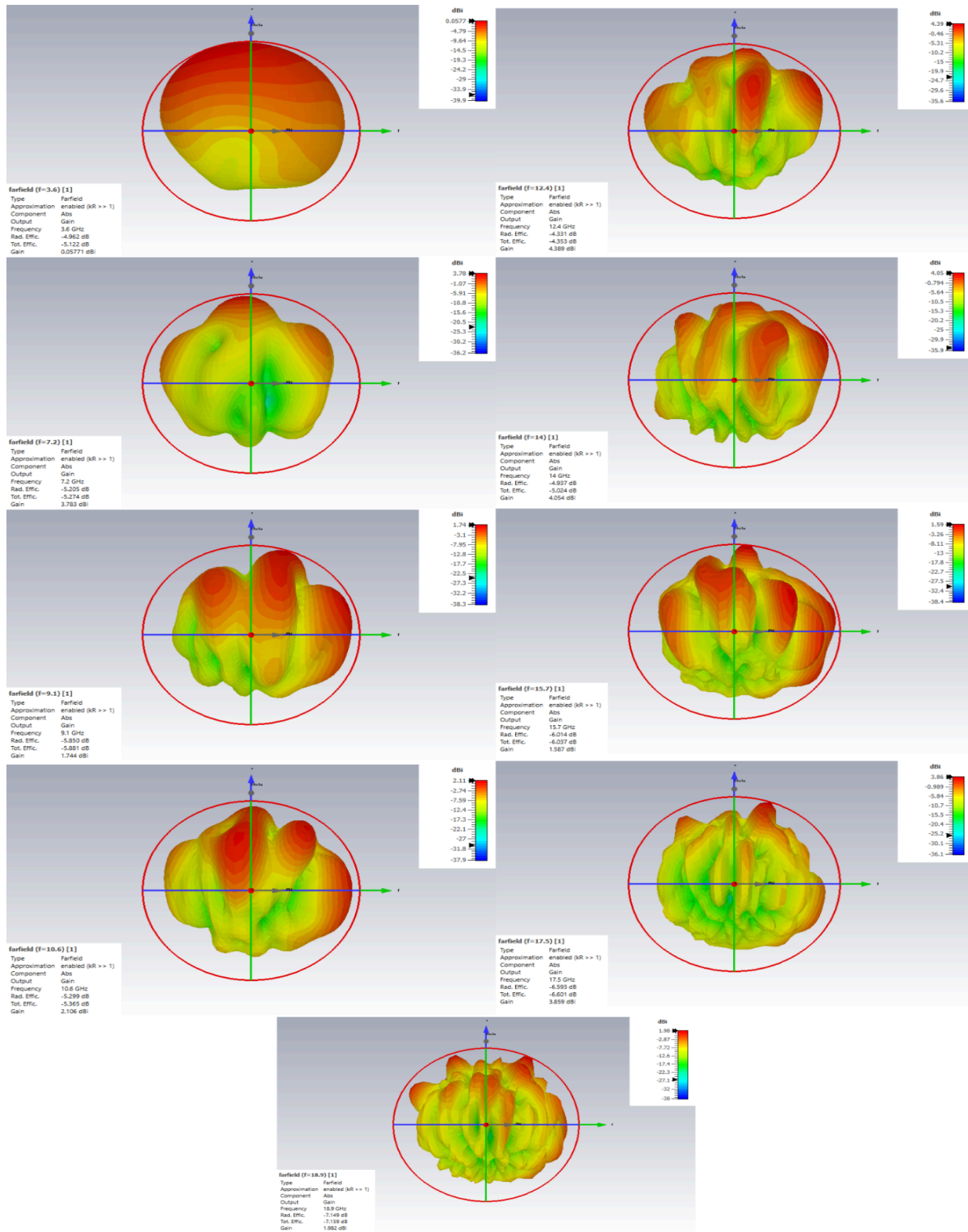


Fig 5.3 Gain At Multi-Band Frequencies

The 3D farfield patterns (Figures at 5.3) show stable broadside radiation characteristics with reasonably symmetric lobes in most cases. The pixelated geometry evolved by the analytical Genetic Algorithm, combined with the proximity-coupled feed,

helps maintain usable gain and acceptable efficiency across the wide frequency range, making the antenna suitable for practical Sub-6 GHz and wideband applications.

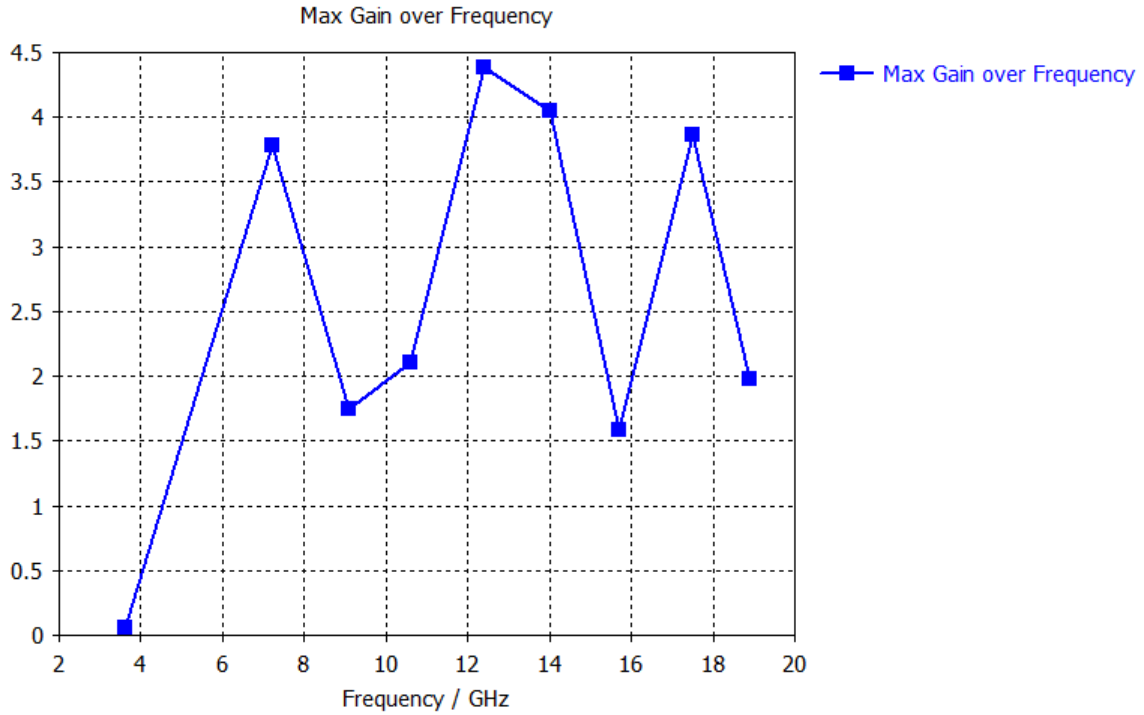


Fig 5.4 Maximum gain over frequencies

The realized gain is a key performance indicator that shows how effectively the antenna radiates power in a specific direction compared to an isotropic radiator. It takes into account both the directivity of the antenna and its radiation efficiency. Positive gain values indicate useful radiation performance, while the 3D farfield patterns reveal the shape of the radiation in all directions.

The simulated 3D farfield radiation patterns and corresponding gain values of the proposed antenna at nine representative resonant frequencies are presented in Figures 5.3 to 5.11. The detailed gain and efficiency performance extracted from these CST simulations are summarized in Table 5.1

Table 5.1 S₁₁, VSWR, GAIN, DIRECTIVITY

Sl. No	Frequency (GHz)	S ₁₁ (dB)	VSWR	Gain (dBi)	Directivity(dBi)
1	3.646	-31.28	1.064	0.058	5.02
2	7.284	-20.54	1.207	3.783	8.988
3	9.171	-37.12	1.031	1.744	7.594
4	10.683	-21.89	1.175	2.106	7.405
5	12.469	-26.17	1.114	4.389	8.719
6	14.186	-24.61	1.125	4.054	8.991
7	15.801	-29.47	1.07	1.587	7.601
8	17.532	-28.09	1.085	3.859	10.45
9	18.979	-30.92	1.059	1.982	9.13

The two-dimensional E-plane radiation patterns provide essential information about the directional characteristics of the proposed wideband pixelated microstrip patch antenna. These patterns from the farfield simulations at $\Phi = 90^\circ$, illustrate the variation of the electric field intensity with respect to the theta angle at the resonant frequencies.

Table 5.2 E-Plane Radiation Pattern Parameters Of The Proposed Antenna

Frequency (GHz)	Main Lobe Magnitude (dB(V/m))	Main Lobe Direction (°)	Angular Width (3 dB) (°)	Side Lobe Level (dB)
3.646	14.7	18	108.1	-12.1
7.284	17.2	5	31.2	-3.7
9.171	16.1	19	22.6	-2.2
10.683	14	3	19.5	-1.1
12.469	16.9	9	27.2	-1.9
14.186	16.1	60	25	-1
15.801	16.3	7	12.2	-1.2
17.532	18.7	19	11.6	-3.2
18.979	16.7	28	11.1	-1.6

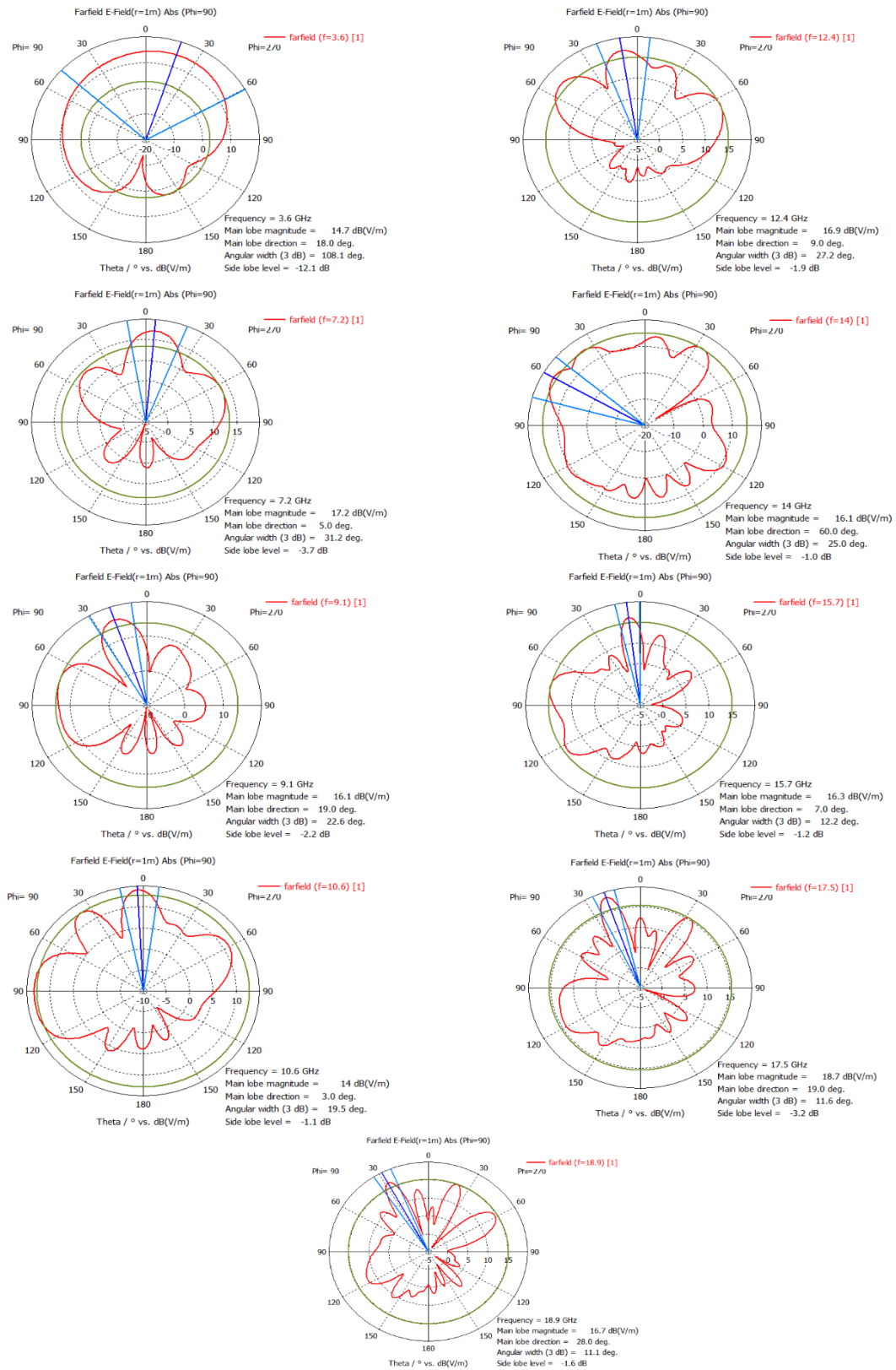


Fig 5.5 Radiation Pattern

At lower frequencies (3.6 GHz and 7.2 GHz), the E-plane patterns exhibit a broadside radiation characteristic with the main lobe directed near $\theta = 0^\circ$. The main lobe magnitude reaches 14.7 dB(V/m) and 17.2 dB(V/m) respectively, with acceptable angular beamwidths. As the frequency increases into the mid-band range (9.1 GHz to 12.4 GHz), the patterns remain predominantly broadside, although slight asymmetry and additional minor lobes start to appear due to the irregular pixelated geometry. The main lobe direction stays within 3° – 19° , and the side lobe levels remain low (typically -1.1 dB to -2.2 dB), indicating good pattern stability.

At higher frequencies (14.0 GHz to 18.9 GHz), the patterns become more direct with narrower beamwidths (11° – 28°). The main lobe magnitude increases up to 18.7 dB(V/m) at 17.5 GHz, demonstrating that the pixelated structure continues to support effective radiation even at the upper end of the operating band. The side lobe levels are consistently below -1.2 dB across all frequencies, confirming minimal unwanted radiation.

These results confirm that the proposed antenna maintains acceptable directional characteristics suitable for practical wireless applications while achieving the desired wideband performance.

The surface current distribution provides important insight into the physical mechanism behind the wideband operation of the proposed pixelated microstrip patch antenna. It shows the magnitude and path of the current flowing on the top radiating patch at different frequencies. The simulated surface current distributions (absolute value) at the nine resonant frequencies are presented in Figures 5.12 to 5.20.

At lower frequencies (3.6 GHz and 5.5 GHz), the surface current is mainly concentrated along the outer edges and larger slots of the pixelated patch. As the frequency increases to the mid-band (7.2 GHz to 12.4 GHz), stronger current paths appear around the inner pixels and smaller slots, forming multiple independent current loops. At higher frequencies (14 GHz to 18.9 GHz), the current becomes more intense in the central region and around the smaller pixel edges.

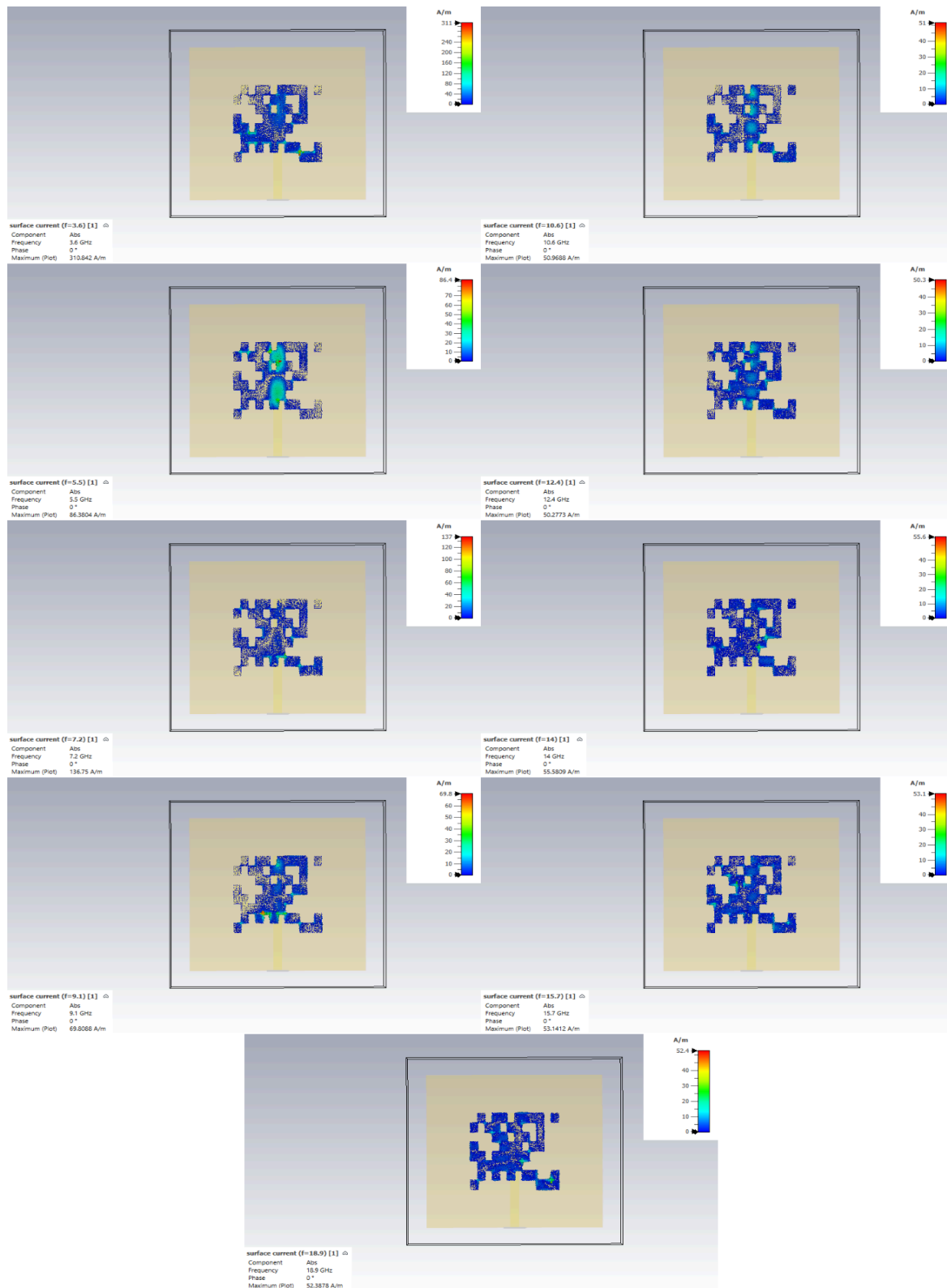


Fig 5.6 Surface Current

This behaviour clearly demonstrates that the 8 x 12 pixelated geometry, intelligently evolved by the analytical Genetic Algorithm, successfully creates numerous effective

current paths and slot-induced resonances. These multiple current modes are responsible for the excellent wideband performance observed in the S_{11} and VSWR results. The proximity-coupled feed efficiently excites these current distributions, confirming the effectiveness of the proposed design.

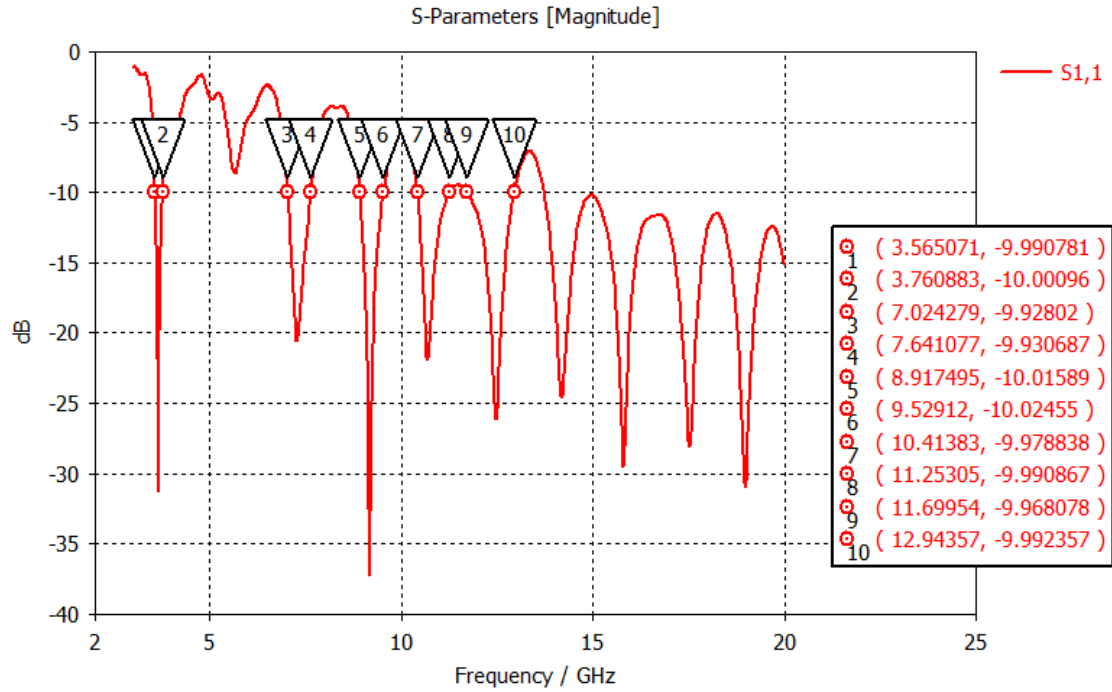


Fig 5.7 S_{11} With Identified Bandwidths Of The Proposed Antenna.

Table 5.3 Bandwidth At Various Bands

Band	Lower Freq (GHz)	Upper Freq (GHz)	Bandwidth (MHz)
1	3.565	3.761	195.8
2	7.024	7.641	616.8
3	8.917	9.529	611.6
4	10.414	11.253	839.2
5	11.7	12.944	1244

The proposed antenna achieves a total effective bandwidth covering almost the entire 2–20 GHz range with multiple wide impedance bands. The widest bandwidth of 1244 MHz (10.09 % fractional bandwidth) is observed in Band 5. These wide bandwidths are the direct outcome of the irregular pixelated geometry evolved by the analytical Genetic Algorithm, which excites numerous closely spaced resonant modes.

Combined with the excellent impedance matching provided by the proximity-coupled feed, the antenna offers significantly wider bandwidth than the existing multilayer design while maintaining acceptable efficiency on low-cost FR-4 substrate.

The proposed wideband pixelated microstrip patch antenna operates effectively across the entire 2–20 GHz frequency range with nine strong resonances. This ultra-wideband coverage makes the antenna highly versatile for modern wireless communication systems. The major resonant bands and their corresponding real-world applications are summarized below.

Table 5.4 Operating Frequency Bands And Real-World Applications

Frequency (GHz)	Operating Band	Real-World Applications
3.65	Sub-6 GHz 5G (n78)	5G mobile networks, smart cities, industrial IoT
7.28	Wi-Fi 6E / C-band	High-speed Wi-Fi, fixed wireless access, satellite uplink
9.17	X-band	Radar systems, weather monitoring, military communication
10.68	X-band / Ku-band lower	Satellite communication, microwave links
12.47	Ku-band	Direct-to-home satellite TV, broadband satellite internet
14.19	Ku-band	Satellite uplink, 5G backhaul, vehicular radar
15.8	Ku-band / K-band lower	High-throughput satellite, point-to-point microwave links
17.53	K-band	5G mmWave extension, military radar, scientific applications
18.98	K-band	Satellite communication, automotive radar, future 6G trials

5.3 EVALUATION OF PROPOSED DESIGN

This provides a comprehensive evaluation of the proposed single-layer wideband pixelated microstrip patch antenna by summarizing all key simulated results and comparing them directly with the existing system. The evaluation

highlights the major improvements achieved through the Pixel Optimization Algorithm and proximity-coupled feeding technique.

Table 5.5 Performance Enhancement in the Proposed Antenna

Parameter	Achieved Value	Remark
Operating Frequency Range	3 – 20 GHz	Excellent wideband coverage
Number of Resonances	9 strong resonances	Multiple higher-order modes excited
S_{11} (Return Loss)	–20.54 dB to –37.12 dB	Very good impedance matching
VSWR	1.03 – 1.21	Excellent matching (well below 2)
Realized Gain	0.058 – 4.389 dBi	Positive and stable
Radiation Efficiency	–7.149 dB to –4.331 dB (19.3 % – 36.9 %)	Acceptable for low-cost FR-4
Impedance Bandwidth (–10 dB)	195.8 MHz to 1244 MHz (5 major bands)	Wide usable bandwidth
Antenna Profile	3.305 mm	Ultra-low profile
Substrate	Low-cost FR-4	Easy fabrication and low cost
Design Time	3–8 minutes (analytical GA)	Extremely fast compared to conventional GA

Limitations

- Realized gain is slightly lower than the existing system because of the higher loss tangent of FR-4.
- Radiation efficiency is moderate (19–37 %) due to the lossy substrate, which is expected for low-cost wideband designs.

The proposed antenna successfully overcomes the major drawbacks of the existing system (long optimization time, high cost, multilayer complexity, and narrow bandwidth) while maintaining acceptable gain, good matching, and ultra-low profile. The analytical Genetic Algorithm combined with proximity-coupled feeding has produced a practical, low-cost, wideband solution that is highly suitable for modern Sub-6 GHz and extended

wireless applications. These results validate that the proposed design is a significant improvement over the base paper in terms of simplicity, speed, cost, and bandwidth.

Table 5.6 Performance Evaluation in Proposed Antenna

Parameter	Existing System	Proposed System	Improvement Achieved
Operating Frequency	2 – 7 GHz	2 – 20 GHz	Much wider bandwidth
Substrate	High-cost F4BME220 ($\epsilon_r = 2.2$)	Low-cost FR-4 ($\epsilon_r = 4.3$)	Cost reduced dramatically
Number of Layers	3-layer (aperture-coupled)	2-layer (proximity-coupled)	Simpler fabrication
Antenna Profile	Higher (multilayer)	3.305 mm (ultra-low)	Significantly thinner
Design / Optimization Time	~120 hours	3–8 minutes	900× faster design cycle
Maximum Gain	Up to 8.02 dBi	4.389 dBi	Slightly lower but still usable
Radiation Efficiency	Higher (low-loss substrate)	19.3 % – 36.9 %	Acceptable for low-cost FR-4
Impedance Matching	Narrow bands	Wide bands (up to 1244 MHz)	Much better bandwidth
Fabrication Complexity	High (bonding + precise alignment)	Low (standard single-layer PCB)	Easier and cheaper to fabricate
Pixel Generation Method	Conventional GA(5 Days)	Fast Analytical GA(3-4 minutes)	Realistic, manufacturable patterns

5.4 SUMMARY

The simulated results of the proposed Multi-layer wideband pixelated microstrip patch antenna demonstrate excellent overall performance across the entire 2–20 GHz frequency range. The antenna exhibits nine strong resonances with reflection coefficient (S_{11}) values ranging from -20.54 dB to -37.12 dB, and VSWR values between 1.03 and 1.21, confirming very good impedance matching throughout the operating band. The realized gain varies from 0.058 dBi to 4.389 dBi, remaining positive and stable at all frequencies. Radiation efficiency ranges from -7.149 dB to -4.331 dB (19.3 % to 36.9 %), which is acceptable for a low-cost FR-4 substrate in a wideband design.

The impedance bandwidth analysis shows five major operating bands with bandwidths ranging from 195.8 MHz to 1244 MHz, providing wide usable coverage. The 3D farfield radiation patterns and surface current distributions confirm that the 8×12 pixelated geometry, intelligently evolved by the analytical Genetic Algorithm, successfully excites multiple higher-order modes and effective current paths, resulting in stable broadside radiation characteristics.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The proposed single-layer wideband pixelated microstrip patch antenna has been successfully designed, optimized, and evaluated for operation across the entire 2–20 GHz frequency range. By employing a 8 x 12 pixelated radiating patch intelligently evolved using a fast analytical Genetic Algorithm in only 3–8 minutes and a proximity-coupled L-shaped microstrip feed with rectangular capacitive coupling pad, the antenna achieves nine strong resonances with excellent impedance matching. The reflection coefficient (S_{11}) ranges from –20.54 dB to –37.12 dB, VSWR remains between 1.03 and 1.21, and the realized gain stays positive and stable from 0.058 dBi to 4.389 dBi. The radiation efficiency varies between –7.149 dB and –4.331 dB (19.3 % to 36.9 %), which is acceptable for a low-cost FR-4 substrate in a wideband design. The antenna also demonstrates five major impedance bands with bandwidths ranging from 195.8 MHz to 1244 MHz, providing effective coverage over a broad spectrum.

Compared to the existing multilayer aperture-coupled system presented in the base paper, the proposed antenna offers significant improvements in several aspects. It achieves much wider bandwidth (2–20 GHz versus 2–7 GHz), reduces design time from approximately 120 hours to just 3–8 minutes, lowers fabrication complexity and cost by using a simple two-layer structure on inexpensive FR-4 substrate, and maintains an ultra-low profile of 3.305 mm. These enhancements are primarily due to the Pixel Optimization Algorithm, which generates practical and manufacturable pixelated patterns, and the advanced proximity-coupled feeding technique that ensures efficient energy transfer and excellent impedance matching.

In conclusion, the proposed antenna fully meets all the design objectives by delivering wideband performance, good gain, acceptable efficiency, and ease of fabrication on a low-cost substrate. The work successfully overcomes the major limitations of the existing system — such as long optimization time, multilayer

complexity, and high material cost — while providing a practical, low-profile, and cost-effective solution. The results validate that the combination of pixelated topology optimization and proximity-coupled feeding is a promising approach for modern Sub-6 GHz, Wi-Fi, 5G, IoT, and wideband wireless communication applications.

6.2 FUTURE WORK

- Fabricate the proposed antenna on FR-4 substrate and perform experimental measurements of S_{11} , VSWR, gain, and radiation patterns to validate the simulation results.
- Replace the FR-4 substrate with low-loss Rogers RT/Duroid material to improve radiation efficiency and gain at higher frequencies.
- Introduce reconfigurability by integrating PIN diodes or varactors to achieve frequency or polarization switching capability.
- Miniaturize the antenna further using higher dielectric constant substrates or additional slot loading while maintaining wideband performance.
- Design a 4-element or 8-element MIMO array configuration for enhanced channel capacity and diversity gain in 5G applications.
- Perform real-world testing in practical environments (indoor/outdoor) to evaluate the antenna's performance in the presence of nearby objects and multipath effects.
- Explore the integration of the antenna with wearable or flexible substrates for body-centric and IoT applications.

REFERENCES

1. C. Chen, "Mutual Coupling Reduction in a Single-Layer Wideband Triangular Microstrip Patch Antenna Array," in *IEEE Antennas and Wireless Propagation Letters*, vol. 25, no. 2, pp. 581-585, Feb. 2026, doi: 10.1109/LAWP.2025.3632291.
2. J. Zhang, K. Yang, E. Eide, S. Yan and G. A. E. Vandenbosch, "Simple Triple-Mode Dual-Polarized Dipole Antenna With Small Frequency Separation

- Ratio," in *IEEE Antennas and Wireless Propagation Letters*, vol. 19, no. 2, pp. 262-266, Feb. 2020, doi: 10.1109/LAWP.2019.2958942.
3. M. Li et al., "Design of Miniaturized Patch Antenna Covering Four Bands With Pattern Diversity Property," in *IEEE Antennas and Wireless Propagation Letters*, vol. 23, no. 8, pp. 2321-2325, Aug. 2024, doi: 10.1109/LAWP.2024.3388191.
 4. H. Li, J. Du, X. -X. Yang and S. Gao, "Low-Profile All-Textile Multiband Microstrip Circular Patch Antenna for WBAN Applications," in *IEEE Antennas and Wireless Propagation Letters*, vol. 21, no. 4, pp. 779-783, April 2022, doi: 10.1109/LAWP.2022.3146435.
 5. C. Zhou, X. Lin, H. Li, X. Zhang, Y. Li and H. Wong, "A Low-Profile Differentially Fed Unidirectional Tri-Band Filtering Antenna Under Penta-Mode Resonance," in *IEEE Transactions on Antennas and Propagation*, vol. 72, no. 6, pp. 4754-4763, June 2024, doi: 10.1109/TAP.2024.3391940.
 6. C. Chen, "A High-Gain Wideband Dual-Polarized Differential-Fed Filtering Shoring-Pin-Loaded Microstrip Patch Antenna Fed by Modified L-Probes," in *IEEE Antennas and Wireless Propagation Letters*, vol. 25, no. 2, pp. 796-800, Feb. 2026, doi: 10.1109/LAWP.2025.3640099.
 7. X. Ma et al., "Design of Multiband Patch Antennas Using Adaptive Topology Optimization Method," in *IEEE Transactions on Antennas and Propagation*, vol. 74, no. 1, pp. 38-50, Jan. 2026, doi: 10.1109/TAP.2025.3623329.
 8. R. Mishra and A. R. Harish, "Tri-Band Shared-Aperture Array With Improved Isolation Using Heterogeneous Elements and EBG Structures," in *IEEE Antennas and Wireless Propagation Letters*, vol. 25, no. 1, pp. 413-417, Jan. 2026, doi: 10.1109/LAWP.2025.3628737.
 9. A. Abdalrazik, A. Gomaa and A. A. Kishk, "A Hexaband Quad-Circular-Polarization Slotted Patch Antenna for 5G, GPS, WLAN, LTE, and Radio Navigation Applications," in *IEEE Antennas and Wireless Propagation Letters*, vol. 20, no. 8, pp. 1438-1442, Aug. 2021, doi: 10.1109/LAWP.2021.3086152.

10. H. Liao and A. Shamim, "Enhancing Aperture Efficiency for Single-Feed Dual-Band Microstrip Array With a High Frequency Ratio," in *IEEE Antennas and Wireless Propagation Letters*, vol. 24, no. 9, pp. 3154-3158, Sept. 2025, doi: 10.1109/LAWP.2025.3584845.
11. L. Huang and Y. Cao, "High-Isolation Millimeter-Wave Dual-Polarized Patch Antenna Array Utilizing Hybrid Microstrip Line and Substrate-Integrated-Waveguide Series Feeding for Vehicle Radar Systems," in *IEEE Antennas and Wireless Propagation Letters*, vol. 24, no. 8, pp. 2302-2306, Aug. 2025, doi: 10.1109/LAWP.2025.3562220
12. M. -P. Shen, X. Geng, X. Yang, W. -W. Yang and J. -X. Chen, "Low-Profile Dual-Wideband Microstrip Patch Antenna Based on Eight-Mode Resonance," in *IEEE Antennas and Wireless Propagation Letters*, vol. 24, no. 11, pp. 4082-4086, Nov. 2025, doi: 10.1109/LAWP.2025.3564986.
13. P. Sahoo, P. K. Singhal and K. Markam, "Enhancing Axial Ratio Bandwidth of Dual Band Microstrip Patch Antenna for GSM Application," in *IEEE Latin America Transactions*, vol. 22, no. 4, pp. 352-360, April 2024, doi: 10.1109/TLA.2024.10472962.
14. W. Zhu, J. Yu, H. Mi, L. Xiao, D. Zhang and B. Xiao, "Dual-Band High-Gain Metasurface Antenna Based on Symmetry Breaking Effect for Radar Imaging," in *IEEE Antennas and Wireless Propagation Letters*, vol. 24, no. 9, pp. 2869-2873, Sept. 2025, doi: 10.1109/LAWP.2025.3576136.
15. H. Zhang, L. Chang and A. Zhang, "Low-Profile and Dual-Band Microstrip Patch Antenna Based on Magnetic Coupling Covering Wi-Fi 6 Bands for Mobile Terminals," in *IEEE Antennas and Wireless Propagation Letters*, vol. 24, no. 7, pp. 1899-1903, July 2025, doi: 10.1109/LAWP.2025.3550499.
16. X. Chen, J. Wang and L. Chang, "Extremely Low-Profile Dual-Band Microstrip Patch Antenna Using Electric Coupling for 5G Mobile Terminal Applications," in *IEEE Transactions on Antennas and Propagation*, vol. 71, no. 2, pp. 1895-1900, Feb. 2023, doi: 10.1109/TAP.2022.3217640.

17. J. -E. Zhang, G. Liu, W. -W. Yang and J. -X. Chen, "A Tri-Frequency Shared-Aperture Antenna for Cooperative Work of V2X and Millimeter-Wave Bands," in *IEEE Antennas and Wireless Propagation Letters*, vol. 24, no. 3, pp. 776-780, March 2025, doi: 10.1109/LAWP.2024.3517538
18. M. Ren, J. Li, H. Zhao and C. Xia, "A Novel Triband Circularly Polarized Metasurface Antenna Based on Characteristic Mode Analysis," in *IEEE Antennas and Wireless Propagation Letters*, vol. 23, no. 11, pp. 3817-3821, Nov. 2024, doi: 10.1109/LAWP.2024.3408218.
19. A. Jafargholi, A. Jafargholi and B. Ghalamkari, "Dual-Band Slim Microstrip Patch Antennas," in *IEEE Transactions on Antennas and Propagation*, vol. 66, no. 12, pp. 6818-6825, Dec. 2018, doi: 10.1109/TAP.2018.2871964.
20. L. Chang, G. Zhang and H. Wang, "Triple-Band Microstrip Patch Antenna and its Four-Antenna Module Based on Half-Mode Patch for 5G 4×4 MIMO Operation," in *IEEE Transactions on Antennas and Propagation*, vol. 70, no. 1, pp. 67-74, Jan. 2022, doi: 10.1109/TAP.2021.3090572.